# Blockchain in Precision Agriculture: A Comprehensive Survey of Architectures, Applications, and Challenges

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The integration of blockchain technology with Precision Agriculture (PA) presents new opportunities for secure, transparent, and decentralized data management in modern farming systems. While blockchain's role in general agricultural applications has been explored in prior surveys, a focused and systematic analysis of its integration within PA—characterized by its reliance on real-time, high-resolution data for decision-making—remains lacking. This survey addresses that gap by reviewing 80 peer-reviewed studies published up to August 2024, with a specific emphasis on the use of blockchain in enabling key PA functions such as monitoring, control, and prediction. We categorize existing applications based on agricultural tasks, identify dominant blockchain platforms (e.g., Ethereum, Hyperledger, IOTA), and analyze their integration with enabling technologies including IoT, edge computing, and Artificial Intelligence (AI). Furthermore, we examine system architectures, consensus mechanisms, and data handling strategies, and assess the evaluation methodologies employed across studies. Based on this analysis, we propose a multi-dimensional taxonomy that captures the functional, technical, and architectural characteristics of blockchain-enabled PA systems. Our findings reveal that while blockchain offers significant advantages in data integrity, automation, and trust-building, its adoption in PA is still hindered by challenges related to interoperability, scalability, energy efficiency, and practical deployment. This review contributes a comprehensive and critical foundation for researchers and practitioners, highlighting both the current state-of-the-art and open research directions necessary to advance blockchain's role in precision farming toward sustainable, secure, and autonomous agriculture systems.

**Keywords**—Literature Review; Blockchain; Precision Agriculture

## 1. Introduction

The convergence of advanced digital technologies and agricultural practices has given rise to PA, a data-driven approach that leverages real-time information to optimize farming decisions and resource management. Within this evolving domain, blockchain technology has emerged as a promising solution to address longstanding challenges related to data integrity, transparency, traceability, and decentralized control. While several review studies have examined blockchain's broader applications in agriculture, there is a growing need to systematically assess its role specifically in PA—particularly in contexts that rely on accurate and secure data collection, such as crop monitoring, irrigation, and disease prediction.

Blockchain's distributed architecture offers unique advantages in agricultural environments, including tamper-resistant data storage, secure communication between devices, and the ability to automate processes through smart contracts. These features are especially critical in PA, where the seamless integration of Internet of Things (IoT) devices, edge computing, and data analytics is essential for efficient farm management and sustainability.

Despite the growing body of research on blockchain-enabled agricultural systems, there remains a lack of consolidated insights into how blockchain has been implemented across diverse PA scenarios, what specific applications have been developed, and what benefits and limitations have been observed in real-world

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deployments. This literature review aims to fill this gap by synthesizing findings from 80 peer-reviewed studies published up until August 2024.

This review has five goals: (a) to identify the key drivers behind the adoption of blockchain technology in PA, (b) to classify the types of applications developed, (c) to examine the architecture and technologies of these systems, (d) to analyze the evaluation methods used in the reviewed studies, and (e) to discuss the challenges and limitations of adopting blockchain technology. By achieving these goals, this study provides a comprehensive overview of the current landscape and suggests pathways for future research and development in blockchain-powered precision farming.

**Significance of the study:** The adoption of blockchain technology in PA represents a novel and evolving research frontier, with the potential to transform how agricultural data is collected, verified, and used. While several reviews have explored blockchain applications across general agricultural domains, there is a noticeable lack of systematic studies focused exclusively on the precision farming context—where real-time data, secure device interoperability, and automated decision-making are crucial.

This review distinguishes itself by offering a detailed and structured synthesis of how blockchain has been applied specifically within PA, addressing farming tasks such as monitoring, control, and prediction. In contrast to prior studies that often concentrate on singular aspects—like security, pest control, or access management—this study takes a comprehensive approach, classifying applications according to the farming tasks they support, the technical architectures they utilize, the blockchain platforms they are built on, and the evaluation metrics they report.

Moreover, this study fills gaps in the literature by examining not only what applications exist, but also how and why blockchain is implemented across different PA scenarios. It contributes a deeper understanding of the motivations behind blockchain adoption (e.g., data integrity, trust, automation), and how its integration with IoT, edge/fog computing, and AI technologies enhances the performance and resilience of PA systems.

By analyzing 80 peer-reviewed studies, this review provides a more granular and holistic mapping of the blockchain-PA landscape than previous reviews. It offers valuable insights for researchers, developers, and agricultural practitioners looking to design secure, efficient, and scalable solutions for modern precision farming.

**Motivation for conducting a literature review:** While the literature on blockchain in agriculture is growing, existing reviews tend to focus on the sector as a whole or on specific technological aspects. Few studies provide a focused and in-depth analysis of how blockchain is applied specifically within PA. Considering the unique technical requirements and complexity of PA systems—where interoperability, real-time responsiveness, and trustworthy data exchange are essential—there is a pressing need for a targeted review. This literature review addresses that gap by synthesizing peer-reviewed studies with a specific focus on blockchain implementations in PA, aiming to clarify current trends, identify best practices, and uncover critical research challenges.

**Highlights of the study:**

- A total of 80 peer-reviewed studies published up to August 2024 were systematically analyzed, focusing exclusively on blockchain use in PA.
- The study categorizes motivations for blockchain adoption, such as security, scalability, data integrity, automation, and trust.
- It identifies and classifies applications developed using blockchain across various agricultural tasks, including monitoring, control, and prediction.
- A detailed examination of system architectures and technology stacks is provided, covering physical, edge, service, and application layers.

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- The review also investigates blockchain platforms used (e.g., Ethereum, Hyperledger, IOTA), evaluation metrics, and reported benefits and challenges.
- Based on the findings, the study outlines future research directions, emphasizing areas such as interoperability, lightweight architectures, and sustainability.

The rest of the paper is organized as follows: Section 2 presents relevant background information and reviews of related literature. Section 3 explores the motivations driving the adoption of blockchain in PA. Section 4 analyzes blockchain-enabled applications, while Section 5 discusses their architectural components. Sections 6 through 9 describe the technologies that make up each architectural layer. Section 10 focuses on the evaluation methods and metrics employed in the reviewed studies. Section 11 outlines the synthesis of the findings, the limitations, and the directions for future research. Finally, Section 12 presents the conclusions that summarize the key insights of the survey.

## 2. Background

In this section, we present the foundational concepts and prior research that contextualize the use of blockchain in PA. We begin by outlining the fundamental characteristics of blockchain technology and its potential advantages in the agricultural domain, particularly in supporting secure, transparent, and decentralized systems. We then review existing literature surveys that have examined blockchain applications in agriculture, identifying their focus areas, contributions, and limitations. This background sets the stage for our focused analysis by highlighting the need for a targeted review that explores how blockchain supports core PA functions such as monitoring, automation, and prediction.

### 2.1 Blockchain Technology

Blockchain is a decentralized, tamper-proof digital ledger that records transactions across a distributed network of computers. It employs cryptographic techniques to ensure data integrity, transparency, and security, making it an appealing solution for domains where trust and reliability are essential. In agriculture—and particularly in PA—blockchain provides a means to manage and validate large volumes of data generated by sensors, devices, and autonomous systems. Its decentralized nature supports secure and transparent data sharing between stakeholders, while smart contracts enable automated decision-making and process enforcement without the need for intermediaries.

In the PA context, blockchain plays a critical role in addressing long-standing issues such as data manipulation, unauthorized access, and lack of trust among parties. Its integration with technologies like the IoT, edge computing, and AI further enhances its ability to support real-time, data-driven agricultural operations. As agriculture continues to evolve, the concept of Agriculture 5.0 is emerging as a vision of the future—where intelligent, interconnected systems drive autonomous decision-making and ultra-precision farming. Unlike previous generations that focused on mechanization (Agriculture 3.0) and digitalization (Agriculture 4.0), Agriculture 5.0 emphasizes the convergence of AI, IoT, robotics, edge computing, and blockchain to enable fully autonomous, sustainable, and resilient farming systems [1]. In this context, blockchain is expected to play a critical role in ensuring the transparency, security, and interoperability of these increasingly complex agricultural ecosystems. It will support real-time trust building among heterogeneous agents—such as drones, autonomous tractors, robots, and cloud services—by offering verifiable data provenance, decentralized identity, and smart contract-based coordination. As PA moves toward Agriculture 5.0, the widespread deployment of blockchain-enabled infrastructures will be essential for unlocking next-generation capabilities in predictive analytics, closed-loop automation, and decentralized supply chains.

### 2.2 Overview of Literature Reviews in Blockchain Technology

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Several literature reviews have explored blockchain applications in agriculture, but most offer limited coverage of PA tasks. Hidayat & Mahardiko [2] focus narrowly on pest control in rice farming, without broader task or platform analysis. Ferrag et al. [3] emphasize IoT security but treat blockchain only as a supporting element. Liu et al. [4] identify general Information and Communications Technology (ICT) and blockchain trends in agriculture, yet lack PA-specific task classification or architectural insights. In a different direction, Dutta et al. [5] briefly mention blockchain in multi-robot PA systems, without discussing implementation. Sendros et al. [6] provide a broad scoping review on blockchain in agriculture, but with minimal focus on PA operations. Akella et al. [7] examine adoption factors in smart agriculture, emphasizing socio-technical issues over task-level applications.

Overall, these prior reviews collectively address valuable subdomains of PA such as pest control, access control, and security, but do not systematically map blockchain applications across the broader range of PA tasks. Moreover, there is a lack of comparative analysis on the types of blockchain used, consensus mechanisms, integration with IoT and AI technologies, or the actual evaluation metrics employed to assess impact.

This review aims to fill that gap by:

- Classifying blockchain-enabled PA applications by the specific tasks they support,
- Detailing the platforms, consensus methods, and data types involved,
- Analyzing performance, benefits, and challenges,
- Highlighting evaluation methodologies used across studies.

By doing so, this paper provides a more holistic understanding of how blockchain technology is being deployed in the PA domain.

Table 1 summarizes the comparative coverage of existing literature reviews and our study. As shown, prior works tend to focus on narrow subdomains or offer high-level discussions, whereas our review delivers a detailed, task-specific, and architectural analysis of blockchain applications in PA.

*Table 1. Review comparison*

| Review Study | Focus Area |
|---|---|
| Hidayat & Mahardiko [2] | Pest detection in rice farming |
| Ferrag et al. [3] | Green IoT security in agriculture |
| Liu et al. [4] | ICT and blockchain for agricultural development |
| Dutta et al. [5] | Multi-robot systems in PA |
| Sendros et al. [6] | General agriculture (PRISMA-ScR) |
| Akella et al. [7] | Blockchain adoption in sustainable farming |
| This Review | PA across all tasks |

## 3. Motivations behind Blockchain deployment in Precision Agriculture

This section discusses the key drivers behind blockchain adoption in the PA domain. Blockchain is mostly known for its relation to ensuring security and transparency in transactions. Identifying other reasons behind blockchain adoption can help us gain insights into the potential of blockchain to transform agricultural processes. The drivers reflecting the multifaceted role of blockchain in addressing challenges and advancing capabilities within PA are summarized in Figure 1.

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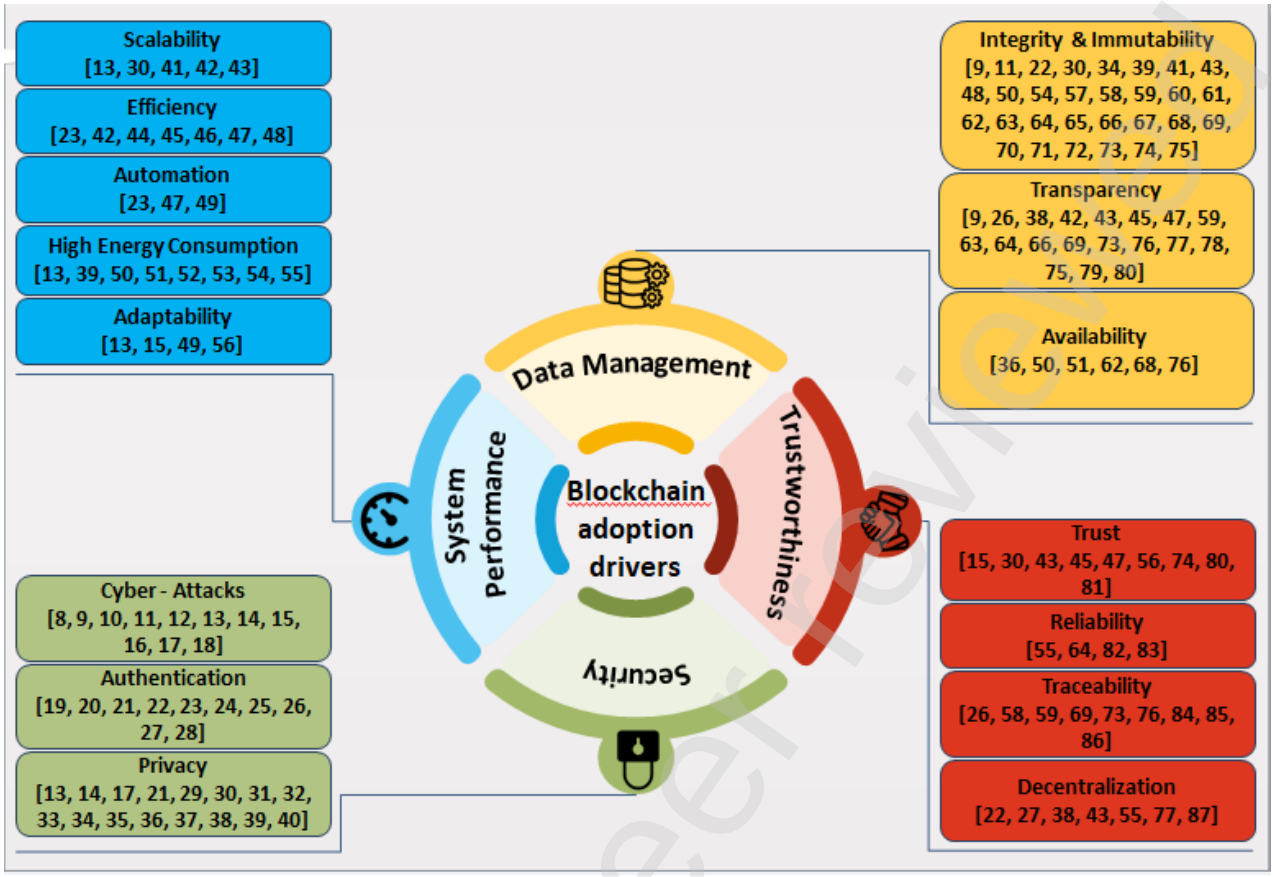


***Figure 1.*** Blockchain adoption drivers

### 3.1 Security & Privacy

Among the challenges that are consistently identified as key motivations for adopting blockchain technology are security and privacy challenges including cyber-attack prevention and effective device authentication.

**Cyber-attacks:** PA systems are increasingly targeted by sophisticated cyber-attacks such as *data poisoning*, *ransomware exploits*, *unauthorized access*, and *distributed denial-of-service (DDoS) attacks*, which threaten the integrity, confidentiality, and availability of critical agricultural operations and data. To address these challenges, blockchain technology, with its decentralized architecture, cryptographic security, and tamper-proof data storage, has emerged as a robust solution. Various frameworks illustrate blockchain's effectiveness in countering such threats: For instance, systems in [8] and [9] focus on preventing *data poisoning* through blockchain-integrated IoT frameworks and real-time anomaly detection. *Attacks on communication integrity*, including *spoofing* and *impersonation*, are tackled by Saba et al. [10], which uses Machine Learning (ML) and blockchain to monitor IoT data reliability. Likewise, *collusion attacks, and malicious data mining attacks* are mitigated in Mishra et al.'s framework [11] using advanced cryptographic mechanisms for secure data transmission. Solutions like Padhy et al.'s [12] and Bapatla et al.'s [13] address *DDoS prevention* through blockchain integrated with fog computing, software-defined networking (SDN), and distributed ledger technologies, improving system resilience while maintaining low latency. Hyperledger-based solutions, including those by Arshad et al. [14] and Smirnov & Teslya [15], provide *secure access control* and context-aware decision-making for agricultural monitoring and autonomous farming robots. Additional innovations, such as Mahalingam & Sharma's hybrid cryptographic models [16], and privacy-preserving aggregation schemes [17], [18], ensure data confidentiality, scalability, and resilience to *advanced cyber threats*.

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**Authentication mechanisms:** Authentication mechanisms refer to the process of verifying the identity of a device or user to ensure that only authorized devices are allowed to access certain resources or perform specific actions. Blockchain-based **authentication** mechanisms in PA verify user identities and devices using cryptographic keys and digital signatures, preventing unauthorized access to sensitive data and systems. Ortega et al. [19] and Navdeep et al. [20] explore how blockchain prevents unauthorized modifications to sensitive agricultural data during collection and transmission. Sharma & Villányi [21] and Quafiq et al. [22] emphasize energy efficiency and secure data processing: the former employs *proof-of-stake consensus* and *lightweight cryptography for device authentication*, while the latter integrates blockchain with Big Data analytics and *Practical Byzantine Fault Tolerance (PBFT)* consensus to enhance scalability and resource efficiency. *Smart contracts* are used by Abijaude et al. [23], Jamil et al. [24] and Jouini & Sethom [25] for device authentication, as well as by Vangala et al. [26] who use them along with PBFT consensus mechanisms for secure Device-to-Device (D2D) and Device-to-Gateway (D2G) authentication. Luo et al. [27] discuss cross-domain authentication in agricultural IoT systems, while Alam et al. [28] focus on securing data transactions with authentication layers.

**Privacy:** Privacy in PA has to do with the way data are handled to ensure that data are used and shared according to the pre-determined agreements. In the realm of data aggregation and collection, several blockchain-based systems have been proposed to address privacy concerns. Decentralized trust mechanisms enable secure data handling in smart farming systems, ensuring that data privacy is preserved throughout the process [29], [30], and without depending on third parties [17]. Chaganti et al. [31] introduce a blockchain-based, cloud-enabled security monitoring framework designed to securely store security anomaly data. Additionally, real-time anomaly detection frameworks leverage blockchain for data aggregation, providing monitoring capabilities while maintaining robust privacy protections [32]. Other systems discuss the integration of blockchain with IoT, highlighting encryption standards to safeguard sensitive farming information [33] and enhance data sharing [34]. Additionally, various approaches have been proposed to enhance data privacy using cryptographic techniques and blockchain technology. These include implementations leveraging an Ethereum Merge-based system [21], InterPlanetary File System (IPFS) [35], Hyperledger Fabric [36], private blockchain-based access control mechanisms [14], smart contracts [37], [38], lightweight blockchain frameworks [39], and IOTA's Tangle technology [13], [40].

### 3.2 System Performance

Blockchain enhances agricultural systems by improving **scalability, efficiency, and automation**. Key motivations for adoption include **high energy consumption and adaptability**, all contributing to system performance.

**Scalability:** As stakeholders, devices, and data points grow in PA, blockchain offers a **scalable** infrastructure for managing transactions efficiently. Noor et al. [41] highlight its necessity in smart farming, as traditional centralized systems struggle with expanding IoT ecosystems. Rahman et al. [30] propose a scalable blockchain-based data-sharing framework for agriculture, ensuring seamless data handling. Bapatla et al. [13] identify scalability as crucial in their blockchain crop monitoring system, while Pullo et al. [42] leverage IOTA's Tangle for high-frequency data transactions. Chacko et al. [43] emphasize blockchain's role in decentralized, large-scale IoT deployments.

**Efficiency** is another key driver of blockchain adoption, optimizing data sharing, verification, and resource utilization—critical for real-time applications like precision irrigation and pest monitoring. Zhao et al. [44] report blockchain-enabled IoT data storage with 10x faster retrieval speeds. Abijaude et al. [23] enhance cocoa bean fermentation monitoring, improving resource use and reliability. Sangeetha et al. [45] and Chang et al. [46] explore blockchain's role in crop monitoring and water management, minimizing inefficiencies. Enescu et al. [47] integrate blockchain with photovoltaic-powered irrigation, improving

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automation and energy efficiency. Pullo et al. [42] demonstrate blockchain-IoT integration reducing water and energy consumption in rice farming. Finally, Vangipuram et al. [48] propose Blockchain-as-a-Service (BaaS) for weather data management, enhancing PA efficiency.

**Automation** plays a critical role as a driver in adopting blockchain technology within various systems in PA. In Abijaude et al.'s solution [23], IoT sensors automate temperature and pH monitoring, ensuring traceability. Enescu et al. [47] integrated irrigation with photovoltaic energy, using blockchain-based automation and smart contracts to optimize resource use. Additionally, robotic coalitions for field tasks leverage blockchain-enabled smart contracts for precision and efficiency [49].

**High energy consumption** is a key driver for blockchain adoption. While blockchain technology is energy-intensive, advancements like proof-of-stake (PoS) and energy-efficient consensus mechanisms reduce its environmental impact, making it viable for sustainable farming. Vangipuram et al. [50] note that conventional IoT and centralized storage are energy-intensive, but integrating blockchain with distributed storage improves efficiency. Awan et al. [51] highlight blockchain's role in minimizing IoT energy use by reducing redundant data transmission and optimizing routing. Krishna et al. [52] emphasize efficient IoT sensor data management, extending network lifespan. Anand et al. [53] and Rao et al. [54] discuss blockchain-enhanced energy-efficient routing in IoT-based sensor networks for PA. Further improvements include Hybrid Blockchain-Ant Lion Optimization (HB-ALO) algorithms in precision farming [55], lightweight blockchain architectures [39], and real-time crop monitoring data sharing [13] for reducing energy consumption.

**Adaptability:** Blockchain's flexible architecture integrates seamlessly with evolving agricultural technologies like IoT sensors, drones, and AI-based analytics, ensuring system **adaptability**. Dawaliby et al. [56] emphasize blockchain's decentralized management, enabling drones to handle IoT malfunctions efficiently, especially in rural areas where human intervention is costly. Smirnov & Teslya [49] and [15] show how smart contracts enhance coordination among robotic coalitions, supporting real-time decision-making and task redistribution. Bapatla et al. [13] discuss IOTA Tangle's scalable, tamper-proof framework, allowing systems to adjust dynamically to environmental changes, optimizing resources in real time.

### 3.3 Data Management

Data **integrity, immutability, transparency, and availability**, which we have collectively categorized under the broader theme of data management challenges, reflect the role of blockchain in ensuring the efficient handling of data within various systems and applications:

**Data integrity & immutability:** Ensuring data **integrity** is a key motivation for adopting blockchain technology in PA, as it guarantees the accuracy and consistency of records throughout their lifecycle. Across various applications, blockchain prevents unauthorized modifications by ensuring that data remain immutable [34], [57], [58], [59]. Feng [60] combines edge computing and blockchain for tamper-resistant storage of agricultural images. Similarly, Arsyad et al. [61] propose the Encapsulating Block Mesh (EBM) framework, which securely tracks crop production transactions using encryption and validation mechanisms. Other researchers highlight the use of Ethereum smart contracts [50], [54], [62], [63], [64], serverless computing [65], and immutable blockchain storage [66] to authenticate sensor data, automate field operations, and safeguard environmental monitoring records. Further applications implement integrity-preserving solutions using IPFS [30], [67], [68], [69] while Sumarudin et al. [70] and Bhattacharya & Dey [71] describe blockchain's ability to securely log data from IoT devices. Mishra et al. [11] introduce a redactable blockchain framework with fog computing to secure data aggregation and prevent unauthorized manipulation. Other studies [22], [39], explore lightweight blockchain frameworks to improve data integrity in edge computing and big data applications. Similarly, Aliyu et al. [9] introduce a blockchain-based security framework aimed at preserving data integrity by detecting anomalies and preventing poisoning attacks in

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real-time.

Blockchain's tamper-proof nature ensures that once data is recorded, it cannot be altered or deleted. Several articles discuss how blockchain ensures data **immutability** in agriculture. Noor et al. [41] introduce a decentralized access control framework to secure IoT devices in smart farming, while Palacios et al. [65] propose a serverless farm management system utilizing Hyperledger Fabric and smart contracts. Jadav et al. [72] integrate AI with Hyperledger, and Mancer et al. [68] employ Ethereum and IPFS to prevent data manipulation. Frikha et al. [73] develop Ethereum-based smart contracts to ensure immutable records of environmental conditions. Chacko et al. [43] combine IoT and blockchain to enhance agricultural operations' data immutability, whereas Sonavane et al. [74] utilize a bioinspired Q-learning model with customized shards and a Proof-of-Performance (POP) consensus mechanism to ensure transaction immutability in Agriculture Internet of Things (AIoT) deployments. Ahmed et al. [75] propose a smart agriculture system using blockchain and implements a cluster head sleep schedule for efficient data aggregation. Finally, Vangipuram et al. [48] introduce W-DaM (Weather Data Management), a BaaS framework, ensuring reliable and tamper-proof climate records for smart agriculture.

**Transparency:** Transparency is also a key driver for blockchain adoption in agriculture, as evidenced by the following studies. Sangeetha et al. [45] highlight blockchain's role in transparent crop quality monitoring, while Enescu et al. [47] show its impact on transparent irrigation systems and resource allocation. Lin et al. [66] demonstrate its role in traceable ICT systems for e-agriculture. Pincheira et al. [76] integrate blockchain with IoT for transparent water management in precision farming. Similarly, Devi et al. [77] emphasize security in IoT-driven agriculture. Vangala et al. [38], [26] explore blockchain-based authentication and smart contracts for reliable data exchange. Ramadona et al. [78] focus on transparency in farming performance evaluation, and Aliyu et al. [9] discuss blockchain's role in securing agricultural data. Elham et al. [79] highlight immutable records for environmental monitoring in poultry farms. Frikha et al. [73] showcase blockchain's use in greenhouse operations, while Chacko et al. [43] emphasize transparency in IoT-agriculture integration. Pullo et al. [42] apply blockchain for sustainable rice farming, and Ahmed et al. [75] discuss its role in secure data transmission. Sizan et al. [59] present blockchain-based crop prediction models. Additionally, Sellami et al. [69] explore blockchain and IPFS for protecting sensor data in precision farming. Shen et al. [80] introduce blockchain-assisted federated learning for transparent data sharing. Vasireddy et al. [63] highlight blockchain's role in irrigation management, and Munim & Islam [64] integrate IoT and blockchain for transparent vertical farming.

**Availability:** The decentralized nature of blockchain ensures high data **availability** and accessibility, even in the face of network disruptions. This is critical for applications like remote monitoring and decision-making in PA. Vangipuram et al. [50] stress its role in addressing latency and centralization in groundwater data management. Sakthi & DafniRose [36] highlight how blockchain ensures data availability, accessibility, and privacy for better decision-making. Alqarni et al. [62] present a blockchain-based framework for secure data exchange between drones and IoT sensors, crucial for smart farming. Awan et al. [51] link availability to robust IoT networks using blockchain for secure data routing. Pincheira et al. [76] emphasize cost-effective IoT devices as reliable data sources for permissionless blockchains in water management. Lastly, Mancer et al. [68] propose a blockchain-IPFS framework to secure and store agricultural data, ensuring availability and integrity.

### 3.4 Trustworthiness

Blockchain fosters trust among stakeholders in the agricultural ecosystem by providing reliable, verifiable, and traceable systems. We grouped the motivations of **Trust, Reliability, Traceability and Decentralization** under the broader category of Trustworthiness to highlight their interconnected nature.

**Trust:** Blockchain enhances stakeholder trust by providing verified, immutable records, ensuring data

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authenticity and transaction integrity—crucial for farmer-supplier-buyer partnerships. Sangeetha et al. [45] showed blockchain's role in IoT-based crop quality monitoring, eliminating intermediaries, enabling direct farmer-distributor connections, and ensuring transparency. Dawaliby et al. [56] integrated blockchain into Unmanned Aerial Vehicles (UAV)-IoT operations, securing drone communication for agricultural tasks. Enescu et al. [47] applied blockchain in photovoltaic-powered irrigation, emphasizing trust in decentralized energy-water trading. Furthermore, Zeng et al. [81] highlighted blockchain's role in secure, community-driven smart water management. Rahman et al. [30] stressed its importance in fostering trust within scalable smart agriculture data sharing systems. Smirnov and Teslya [15] exploited blockchain for trusted robotic coalition interactions in PA. Sonavane et al. [74] demonstrated blockchain -based AIoT trust metrics optimizing security and service quality. In addition, Chacko et al. [43] explored blockchain -integrated IoT (BIoT), enhancing transparency, accountability, and resilience. Shen et al. [80] proposed a blockchain -assisted federated learning framework to ensure trust, privacy, and improved pest-disease management models.

**Reliability**: Blockchain enhances reliability in agriculture by ensuring uninterrupted operations and minimizing downtime. Wu & Tsai [82] integrate IoT and blockchain to eliminate single points of failure, improving system reliability. Devi et al. [83] emphasize blockchain's role in securing data reliability, crucial for tasks like maize disease detection and classification accuracy. Rajeshkumar and Sureshkumar [55] use HB-ALO to enhance communication reliability and secure routing in PA. Munim & Islam [64] highlight blockchain's immutable ledger in vertical farming, ensuring secure storage and transparent data retrieval, fostering stakeholder trust.

**Traceability:** Blockchain also ensures precise traceability of agricultural products by maintaining an immutable record of transactions and events. This meets regulations and helps consumers verify product origins and certifications. Pincheira et al. [76] integrate IoT with blockchain for water management, ensuring data integrity and incentivizing traceable smart contracts. Vangala et al. [26] propose a blockchain-enabled smart farming model that improves data auditability and transparency. Widayat & Köppen [84] highlight blockchain's role in securing food provenance by integrating IoT to store sensor and camera data securely. Frikha et al. [73] apply blockchain and DL for ensuring supply chain traceability. Nguyen et al. [85] explore blockchain's role in sustainable coffee value chains in Vietnam, fostering trust among stakeholders. Dewangan and Vij [58] emphasize livestock traceability using IoT-based data and blockchain-secured records. Kataev et al. [86] discuss blockchain's trust-building capabilities through secure data management. Sizan et al. [59] introduce a framework combining IoT, ML, and blockchain for PA, ensuring data integrity and transparency. Sellami et al. [69] develop a blockchain-IoT prototype for decentralized data storage and sensor traceability in smart farming.

**Decentralization:** Decentralization is a key driver for adopting blockchain in agriculture, addressing inefficiencies and vulnerabilities of centralized systems. Several studies emphasize its transformative potential. Shyamala Devi et al. [77] and Song & Li [87] highlight how decentralization eliminates third-party intermediaries, enhancing transparency and securing data, particularly in remote farming environments. Similarly, Rajeshkumar & Sureshkumar [55] demonstrate the role of hybrid blockchains in improving communication reliability and energy-efficient data transfer. Vangala et al. [38] and Luo et al. [27] focus on efficient resource management and secure communication in agricultural IoT systems, with decentralization mitigating single points of failure and supporting lightweight authentication. Chacko et al. [43] add that decentralization enhances scalability and reliability in IoT-blockchain integrated systems, addressing bottlenecks and improving performance. Moreover, Quafiq et al. [22] explore integrating blockchain with Big Data architectures to decentralize storage and decision-making, enabling precision farming. These studies collectively highlight decentralization's role in enhancing security, scalability, and cross-domain collaboration, driving blockchain adoption for sustainable and smart agriculture.

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## 4. Applications developed with blockchain technology in the PA domain

This section investigates the types of blockchain-driven applications developed to support PA. By analyzing findings from relevant studies, we aim to uncover the specific farming tasks these applications enable, the role of blockchain in facilitating these tasks, and the types of crops or farms where blockchain technology has been implemented. Through the analysis of the selected studies, we identified three primary categories of agricultural tasks where blockchain technology has been applied: **Monitoring, Control (Automation)/Management, and Prediction**.

Figure 2 provides an overview of the farming tasks that are supported by blockchain applications across various agricultural domains. For each type of agricultural field, we outline the tasks facilitated by blockchain technology.

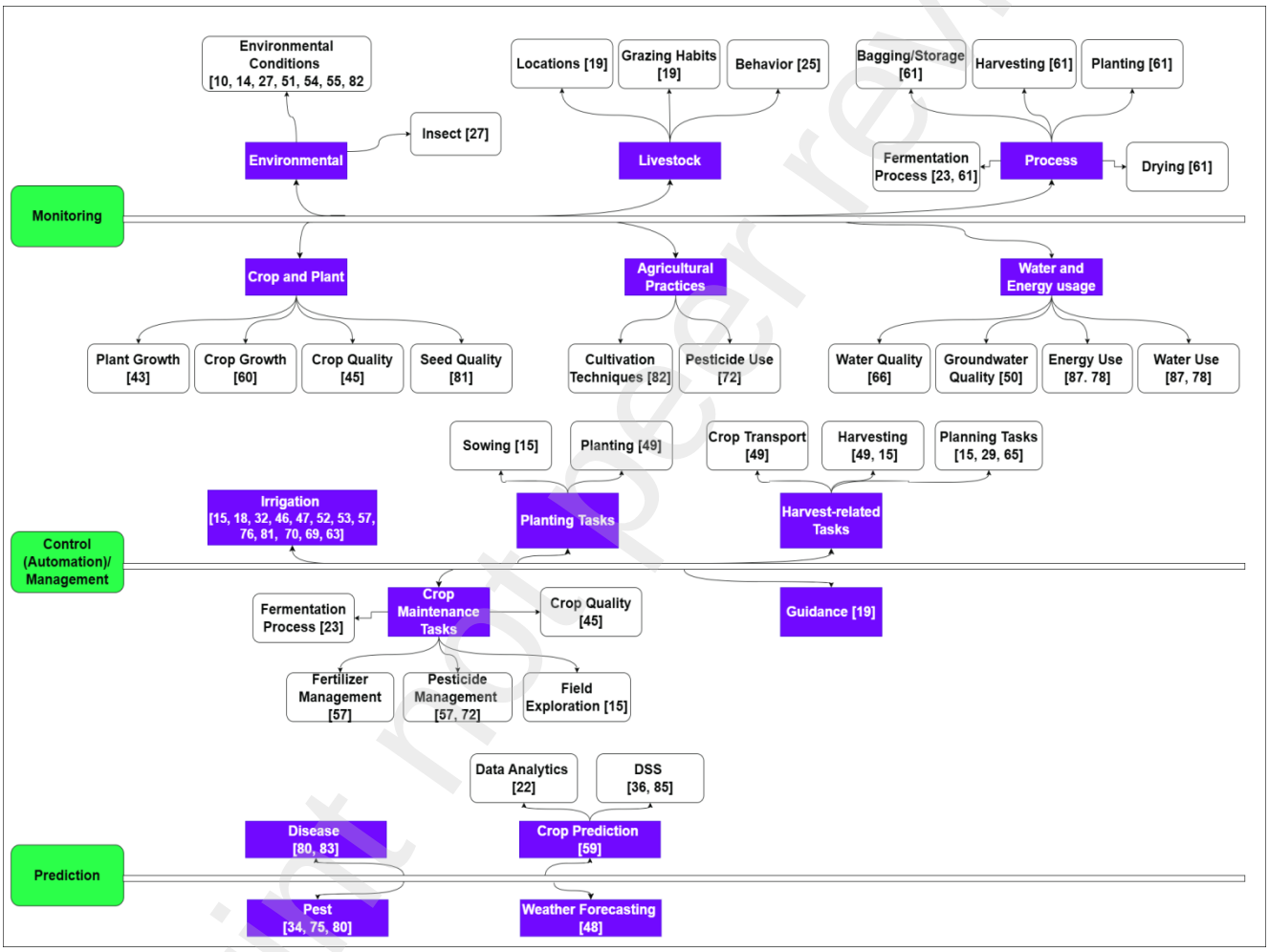


***Figure 2.*** Agriculture tasks supported by applications integrating blockchain technology

### 4.1 Monitoring

Monitoring in agriculture involves the procedure of tracking specific environmental, soil and crop conditions and indices critical for maintaining the health and productivity of agricultural systems. The majority of the articles reviewed, point out that monitoring is the primary focus of the systems demonstrated.

**Environmental monitoring**: Many PA applications focus on deploying systems that monitor environmental conditions utilizing blockchain and IoT technologies. Saba et al. [10] proposed a trust-based decentralized system that combines IoT, blockchain, and ML to automate data collection in monitoring soil and crop conditions. Wu & Tsai [82] developed a private blockchain-based system to monitor climatic

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parameters while Awan et al. [51] introduced an energy-efficient routing protocol for Wireless Sensor Networks (WSNs), utilizing blockchain to enhance data integrity and reliability in environmental monitoring. Arshad et al. [14] designed a scheme to monitor temperature, light, humidity, and soil moisture securely. Rajeshkumar & Sureshkumar [55] proposed an HB-ALO algorithm to minimize energy consumption and ensure secure data transfer in IoT networks for agriculture. Luo et al. [27] presented a hybrid blockchain-based cross-domain authentication scheme, enabling integration of IoT devices across agricultural domains for effective monitoring. Finally, Rao et al. [54] introduced a blockchain-enhanced cluster-based routing protocol to optimize network lifetime and data integrity in agricultural monitoring systems.

**Water and energy usage monitoring:** Song & Li [87] and Ramadona et al. [78] emphasize the use of blockchain-enabled networks to support sustainable farming by monitoring water and energy usage. They incorporate low-power transaction nodes like smart water and electricity meters and relay nodes to enhance data transmission efficiency and security. G-DaM system [50] integrates Distributed Data Storage (DDS) and blockchain technologies to *manage groundwater quality* data effectively. The system addresses challenges such as data tampering, transmission risks, and centralized storage issues. Similarly, Lin et al. [66] developed an e-agriculture system leveraging blockchain technology for water quality monitoring. It highlights a decentralized framework to store and validate water quality data using blockchain's immutable and distributed ledger system.

**Crop, seed and plant monitoring:** Feng [60] integrates blockchain technology into his system to enhance *crop growth monitoring*. He proposes a system architecture that leverages IoT sensors to capture real-time crop images, processes the data using advanced Hash algorithms, and ensures secure storage through blockchain. Similarly, Sangeetha et al. [45] presented IoT-Blockchain integration for *crop quality* classification, emphasizing how sensor data can classify yields into quality grades and ensure secure transactions between farmers and distributors, eliminating intermediaries and promoting fair pricing. In addressing *seed quality,* Zeng et al. [81] introduced a blockchain-based smart water and seed management system, highlighting the critical role of monitoring and preserving seed conditions to boost productivity. Lastly, Chacko et al. [43] investigated IoT-Blockchain architectures for *plant growth* tracking, showcasing how such systems enhance resource optimization and transparency in agricultural practices.

**Livestock monitoring:** Articles focusing on livestock monitoring systems highlight advancements in monitoring animal behaviors and improving farming efficiency. Ortega et al. [19] introduce a decentralized system using UAVs integrated with blockchain and IPFS. This system enables secure tracking of livestock *locations* and *grazing habits*, optimizing farm operations and reducing manual labor. Meanwhile, Jouini & Sethom [25] describe an IoT-based solution leveraging blockchain for monitoring livestock *behavior*. Equipped with connected collars, animals' activities, including feeding and grazing patterns, are recorded and analyzed. The data is transmitted to a farmer's mobile application through a secure, decentralized blockchain system.

**Agriculture processes monitoring:** Arsyad et al. [61] introduce a modular blockchain-based system designed for monitoring and documenting various stages of cocoa production, including *fermentation, bagging/storage, harvesting, planting, and drying.* This system ensures that all farming events, from water usage to the storage of dried cocoa, are traceable and tamper-resistant. Similarly, Abijaude et al. [23] developed a blockchain-IoT-based system specifically for monitoring the *fermentation* process of gourmet cocoa beans. The system integrates IoT sensors to capture critical environmental parameters such as temperature and humidity, which are pivotal in ensuring high-quality fermentation.

**Agriculture practices monitoring**: One of the practices monitored in the reviewed studies is the use pf pesticides. Jadav et al. [72] introduce a blockchain and AI-empowered framework for predicting and managing pesticide use in agriculture. It emphasizes storing pesticide usage data securely on immutable

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blockchain ledgers to prevent manipulation and enhance transparency. The approach also integrates AI models to predict pesticide thresholds and improve agricultural outcomes while minimizing environmental and health impacts. Wu & Tsai [82] highlight the use of blockchain to securely track and store cultivation-related data, such as soil quality and *cultivation techniques*, ensuring data integrity and supporting precision farming initiatives. Luo et al. [27] describe a hybrid blockchain framework that includes specialized *insect* monitoring systems.

### 4.2 Control (Automation)/Management

Control and management systems in PA integrate technology, data analytics, and automation to enable site-specific management of agricultural operations, optimizing resources, reducing labor, and increasing productivity. Automation in *irrigation*, *pest and crop management*, and *nutrient application* enables real-time responses to environmental conditions, which is especially beneficial in regions with labor shortages or challenging climates.

**Irrigation management:** The integration of blockchain into irrigation systems has been widely explored, demonstrating its potential in PA. Hossain et al. [57] propose a smart framework combining IoT, ML, and blockchain for optimized irrigation, focusing on secure data sharing and real-time decisions. Ting et al. [32] enhance this with fuzzy logic and edge computing for intelligent water management. Enescu et al. [47] develop a blockchain-powered irrigation system using photovoltaic energy for small-scale farmers. Pincheira et al. [76] introduce a cost-effective blockchain-IoT solution ensuring water data reliability and incentivizing sustainability through smart contracts. Zeng et al. [81] highlight blockchain's role in real-time water monitoring and resource allocation. Krishna et al. [52] propose a cloud, blockchain, and IoT system for transparent water usage. Chang et al. [46] emphasize blockchain's role in securing and sharing real-time irrigation data, ensuring fault detection. Anand et al. [53] integrate blockchain with energy-efficient algorithms for streamlined irrigation and data integrity. Sumarudin et al. [70] and Vasireddy et al. [63] explore Ethereum blockchain for secure IoT sensor data management, addressing transparency and tampering. Smirnov & Teslya [15] apply blockchain to coordinate robotic irrigation. Jegadeesan et al. [18] propose a lightweight blockchain framework for secure irrigation data aggregation. Sellami et al. [69] combine blockchain and IPFS to protect sensor data, ensuring sustainability in precision irrigation.

**Planting and Harvest-related tasks:** In this category we grouped together agricultural tasks that have to do with harvesting. Patel et al. [29] focus on using blockchain to enhance **crop recommendation** through a recommendation system. Sensors collect soil attributes, such as nitrogen, phosphorus, potassium, pH levels, and moisture. This data is stored on a blockchain, ensuring it is tamper-proof and transparent. The system uses ML algorithms to analyze soil health and suggest suitable crops. Blockchain's role is pivotal in validating sensor data and maintaining trust in the recommendations, enabling farmers to make data-driven crop choices based on real-time conditions. Smirnov & Teslya's studies [15] , [49] illustrate the integration of blockchain and smart contracts—specifically using the Hyperledger Fabric platform—into coalitions of robots in PA. These robots collaborate on tasks such as **soil analysis, crop selection, planting, sowing, harvesting, and crop transport**. Blockchain ensures secure communication and coordination by maintaining a distributed ledger of tasks and their outcomes, while smart contracts enable the autonomous distribution of roles and responsibilities among the robots. Palacios et al. [65] propose an integrated system using serverless functions and blockchain for **farm management**, particularly for task and crop recommendation. By leveraging smart contracts and shared ledgers, the system automates task assignments and manages resource allocation.

**Guidance**: Ortega et al. [19] describe the use of UAVs enhanced with blockchain. Blockchain ensures data integrity and immutability, which aids in managing the guidance of livestock by securely transmitting GPS data and ensuring accurate positioning through a distributed ledger.

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**Crop Maintenance Tasks:** Hossain et al. [57] combine blockchain with IoT and ML models to optimize **fertilizer and pesticide use**. Real-time soil quality and moisture data are processed to recommend the precise type and amount of fertilizer and pesticide required for specific crops, maximizing productivity while minimizing waste. Blockchain ensures that the data regarding chemical application is secure and traceable, promoting sustainable farming practices. Pesticide use is also the focus of Jadav et al.'s study [72], where AI models predict the threshold beyond which pesticide use could harm crop quality and human health. Data on pesticide usage and environmental impact are securely stored in immutable blockchain ledgers. Abijaude et al. [23] employ smart contracts to automate actions such as stirring the cocoa mass based on specific criteria, ensuring consistency and quality in the **fermentation process**. Blockchain enhances transparency, allowing traceability throughout the fermentation cycle. Sangeetha et al. [45] discuss how sensors track environmental parameters such as temperature, humidity, and CO2 levels, in order to use the gathered data to classify crops into **quality grades**. Finally, Smirnov & Teslya [15] use smart contracts to coordinate coalitions of robots for tasks like **field exploration**.

### 4.3 Prediction

Prediction-focused applications leverage historical data and current environmental conditions to anticipate future needs and risks, allowing farmers to take preventive actions.

**Disease prediction:** Devi et al. [83] and Shen et al. [80] developed systems that can help in disease prediction. The first is a residual neural network with decentralized blockchain architecture to detect maize leaf diseases. Blockchain ensures secure and decentralized storage and sharing of disease data, improving reliability and transparency in data handling. The second integrates blockchain technology with federated learning to predict and monitor rice pest and disease outbreaks. Blockchain is preventing unauthorized access during data sharing, through homomorphic encryption and secure aggregation.

**Pest detection:** Rahman et al. [34] introduce a smart framework combining IoT, ML, and blockchain for real-time pest detection in paddy fields. The system utilizes cameras and sensors for capturing data, which is then, processed using DL models like VGG19 for pest identification. Ahmed et al. [75] present an innovative approach to pest attack prevention using blockchain technology. It integrates IoT sensors to monitor pest activity and environmental conditions and employs a "Cluster Head Sleep Schedule" mechanism to optimize energy use while ensuring real-time pest detection.

**Crop Prediction:** According to Sizan et al. [59], blockchain enhances IoT- and ML-driven frameworks for crop prediction by ensuring data transparency, security, and integrity. The system uses IoT devices to collect real-time environmental and soil data, such as temperature, humidity, and nutrient levels, which are then, processed using ML algorithms for precise crop yield forecasting. Two articles demonstrate the application of blockchain technology in **Decision Support Systems (DSS)** for agriculture, providing innovative solutions to enhance data management, security, and sustainability in the agricultural sector. Sakthi & DafniRose [36] propose a blockchain-enabled agricultural knowledge discovery system integrated with edge computing and IoT. The system enhances the smart DSS by securely storing and managing agricultural data collected through sensors. Nguyen et al. [85] created an INNovative and Smart Agriculture (INNSA) platform for Vietnam's coffee industry. The platform integrates IoT, AI, and blockchain to optimize the coffee value chain. Blockchain enhances traceability and transparency in farm operations, while AI supports tasks like disease prediction and resource optimization. Quafiq et al. [22] introduce a **data analytics** system that integrates blockchain technology with Big Data platforms such as Hadoop. The system employs cryptographic techniques alongside PBFT to enhance both data integrity and scalability.

**Weather Forecasting:** Vangipuram et al. [48] introduce the W-DaM system, which integrates BaaS with IoT hubs for weather prediction. This system collects weather data from IoT devices, processes it at edge computing hubs, and stores it on a blockchain ledger to ensure immutability and trustworthiness.

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## 5. The architecture of applications integrating blockchain technology in Precision Agriculture

This section aims to explore the architecture of applications integrating blockchain technology in PA. As PA increasingly relies on data-driven ways to improve productivity and sustainability, the use of blockchain technology provides intriguing options for data integrity, traceability, and decentralized decision making. This section examines the structural architecture and essential components of these applications to shed light on how blockchain interacts with diverse agricultural technologies, such as IoT devices, sensors, and cloud platforms, to promote secure and transparent farming operations. A visual presentation of the architecture is presented in Figure 3.

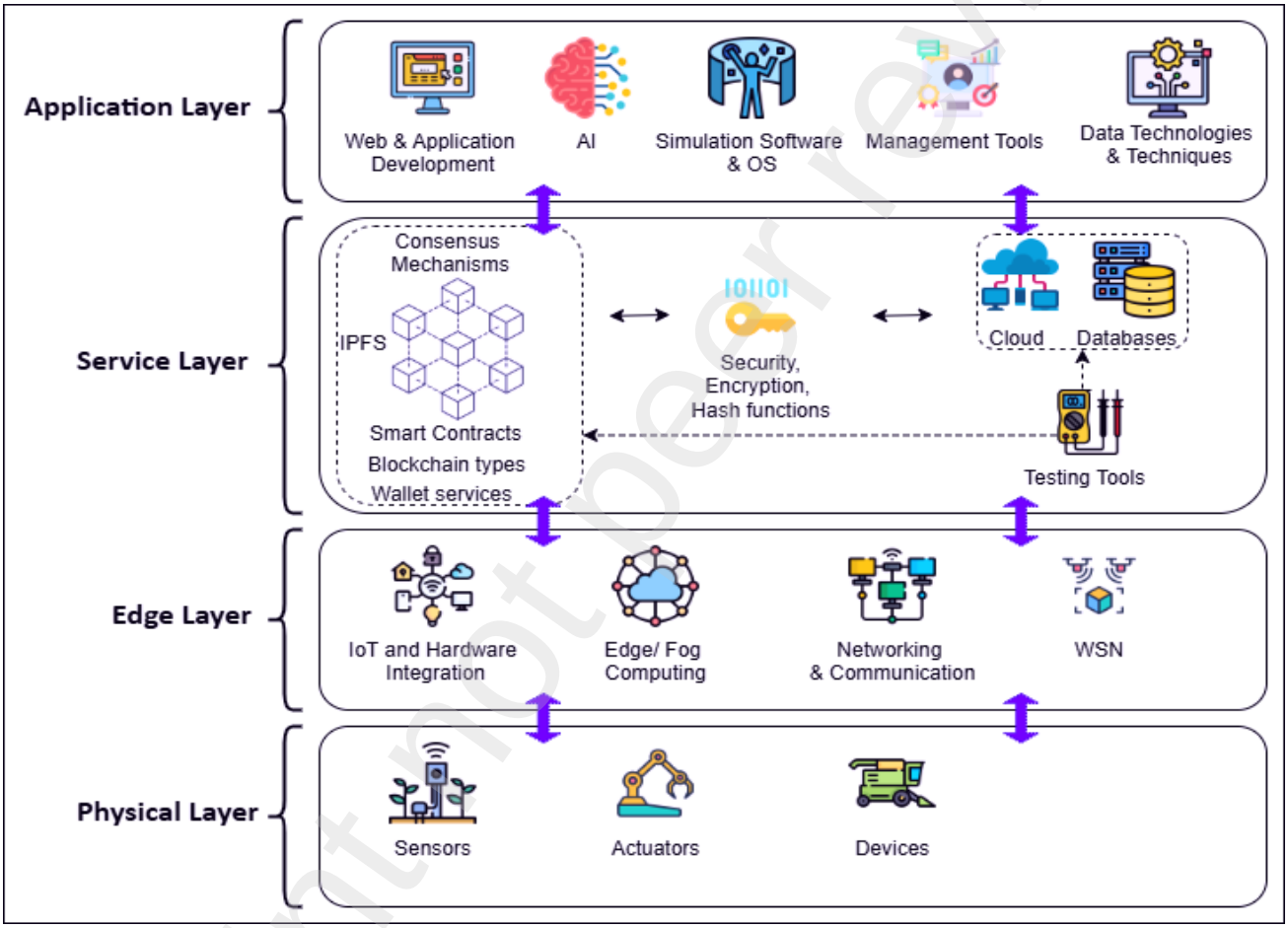


*Figure 3. Blockchain- based architecture in PA*

### A. Physical Layer

This layer includes all hardware components directly involved in sensing and interacting with the physical environment of the farm. It is divided into:

- Sensors (e.g., temperature, humidity, pH, soil moisture) that collect data from the field.

- Actuators (e.g., water pumps, heaters, feeders) that perform actions based on data or control logic.

- Devices (e.g., UAVs, GPS modules, Radio Frequency Identification (RFID) tags, tractors, robots) that support mobility, tracking, or farming operations.

The physical layer is the foundational level where real-world conditions are sensed and acted upon.

### B. Edge Layer

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This layer includes edge and fog computing technologies, networking protocols, and IoT hardware responsible for:

- Real-time data collection and pre-processing close to the source (e.g., microcontrollers like Arduino, ESP32).

- Communication between devices using technologies like Long Range Wide Area Network (LoRaWAN), MQTT, RFID, and 6G.

- Networking and WSNs to ensure low-latency, energy-efficient data transmission and automation.

The Edge Layer acts as the bridge between the physical world and the computational backend.

**C. Service Layer**

This layer integrates blockchain-specific components that provide trust, security, and automation:

- Blockchain Platforms (e.g., Ethereum, Hyperledger, IOTA).

- Smart Contracts that automate agreements and enforce rules.

- Encryption Techniques, Hash Functions, and Consensus Mechanisms (e.g., PoS, PBFT) for secure and immutable data handling.

- Distributed Storage systems like IPFS and wallets (e.g., MetaMask).

It's the core layer that brings decentralization, transparency, and automation into PA applications.

**D. Application Layer**

This is the top-most layer, where user-facing tools and services exist. It includes:

- Web and mobile applications for farmers, suppliers, and consumers.

- AI and simulation software for predictions and decision support systems.

- Management dashboards and monitoring tools to visualize and interact with real-time data.

The Application Layer presents the processed information and blockchain-secured insights in a usable form to stakeholders.

## 6. Physical Layer technologies

In the Physical layer we grouped the devices into three primary categories: ***Sensors, Actuators, and Devices***. *Sensors* collect and transmit critical data, while *Actuators* are responsible for enabling the system to perform actions. All other devices that did not fall into these two categories were grouped under the broader "*Devices*" category.

### 6.1 Sensors

This section includes all of the equipment used to sense and gather different types of useful data for the efficient operation and enhancement of the production process. The various sensor types that have been identified are listed in Table 2.

*Table 2. Sensors used by practitioners*

| Sensors | Description | Articles |
|---|---|---|
| **Humidity (Hygrometer)** | Measure the humidity level in the air. It detects changes in the humidity and converts the data into an electrical signal that can be monitored and recorded.<br>*Examples:* HS200, SY-HS-220, and DHT-11, as well as sensors integrated into kits like Arduino and NodeMCU ESP8266. | [8], [13], [14], [23], [24], [28], [30], [31], [32], [33], [34], [35], [36], [40], [43], [45], [46], [48], [51], [52], [53], [54], [55], [59], [64], [70], [71], [73], [75], [79], [82], [84], [85], [86] |

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| | | |
|---|---|---|
| **Motion/ Sound** | Motion sensors detect physical movement within a designated area. Sound sensors, capture sound waves and convert them into electrical signals, allowing for the detection of sound levels or specific noises. We included accelerometers in this category, as they also detect movement or changes in orientation, complementing the functions of motion and sound sensors. | [8], [9], [36] |
| **Temperature** | All sensors that measure temperature, whether in the air or soil.<br>*Examples*: L-35, LM5.B, SHT15, and DHT-11, as well as temperature sensors integrated into development kits such as Arduino and NodeMCU ESP8266. | [8], [9], [10], [13], [14], [18], [23], [24], [28], [31], [32], [33], [34], [35], [36], [37], [38], [40], [42], [43], [45], [46], [48], [51], [52], [53], [54], [55], [58], [59], [63], [64], [67], [68], [70], [71], [72], [73], [75], [77], [79], [81], [82], [84], [85], [86] |
| **Light** | Light exposure sensors, Light Dependent Resistors (LDR), light intensity sensors<br>*Examples*: BH1750 and BH1750FVI, as well as sensors integrated into development kits, such as Arduino | [8], [9], [14], [30], [31], [32], [33], [35], [57], [64], [70] |
| **Air Pressure** | Also known as barometric pressure sensors, they monitor the atmospheric pressure of gases. They function by detecting the force exerted by the air molecules in a given area and converting this pressure into an electrical signal. | [9], [10], [40], [48], [51], [55], [77], [82] |
| **Gas** | Measure levels of key elements such as Nitrogen (N), oxygen, N2O, and CH4, as well as sensors like the MQ137 for ammonia content and the MG811 for CO2 detection. | [24], [29], [38], [45], [59], [64], [67], [68], [85] |
| **Soil and Nutrients** | Measure phosphorus (P) and potassium (K) levels, as well as sensors that assess soil conditions, electrical conductivity, and other soil-related parameters. | [27], [29], [59], [74], [82], [85] |
| **PH** | Measure the pH levels in water or soil.<br>*Examples*: SEN-00239 pH sensor and ISFET-based sensors (often used for pH measurement due to their ability to detect ion concentrations) | [14], [23], [29], [33], [52], [53], [57], [59], [64], [67], [70], [72], [84], [85] |
| **IoT** | A device that collects data from the physical environment and transmits it over the internet for analysis and monitoring. | [11], [16], [17], [20], [21], [22], [26], [39], [41], [47], [56], [62] |
| **Moisture** | Measure the water content in soil, crops, or other materials. They work by detecting the amount of water present and providing data to help optimize irrigation, improve crop growth, and conserve water.<br>*Examples*: Capacitive and Dielectric Soil Moisture sensors, EC1258, FC-28, EC-5, and Echo20. | [10], [14], [18], [29], [32], [34], [36], [37], [38], [40], [42], [43], [45], [46], [51], [52], [53], [54], [55], [57], [63], [64], [68], [69], [70], [71], [73], [75], [81] |

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| | | |
|---|---|---|
| **Rain** | Detect the presence and amount of rain in an environment. They are commonly used in automated systems like irrigation systems and windshield wipers, helping to trigger actions based on weather conditions. In irrigation, rain sensors can prevent unnecessary watering by detecting rain and signaling the system to turn off. The reviewed articles also mention MH-RD Rain and rainfall sensors. | [18], [36], [38], [40], [57], [59], [70], [72] |
| **Cameras** | Capture high-resolution images and video to monitor crop health, soil conditions, and livestock. These sensors, often integrated with drones or other automated systems, provide valuable data for precision farming. They enable farmers to detect issues like pest infestations, water stress, and nutrient deficiencies early, helping to optimize resource use and improve yields. Camera sensors can also assist in tasks such as crop counting, field mapping, and the analysis of plant growth patterns, making them essential tools for modern agricultural practices. | [12], [32], [37], [73], [84] |
| **GPS** | Use satellite signals to determine the precise location of an object on Earth. By calculating the distance between the sensor and multiple satellites, GPS sensors can provide real-time data about an object's latitude, longitude, and altitude. These sensors are widely used in navigation systems, vehicles, smartphones, and various applications like agriculture, surveying, and geolocation tracking. GPS sensors enable accurate location-based services and are crucial for tasks requiring precise positioning. | [12], [13], [35], [82] |
| **Water** | Measure water quality, water level, water usage or consumption, and parameters such as Electrical Conductivity (EC) and Total Dissolved Solids (TDS). Literature also refers to smart water meters.<br>*Examples*: YF-201 Water Flow Sensor | [30], [36], [37], [42], [47], [50], [51], [55], [64], [66], [67], [76], [78], [81], [87] |
| **Pest Identification** | Detect, monitor, and identify pests that can impact crops, livestock, or ecosystems. This includes devices like Pest Detection Sensors, which use advanced sensing technologies to detect the presence of pests, and Insect Monitoring Systems, which track insect populations and behaviors in real time. | [27], [36], [75] |
| **Environmental Monitoring** | Assess leaf wetness, solar radiation, wind direction and speed, precipitation, as well as weather and meteorological data monitoring systems. | [27], [40], [48], [70] |
| **Smart Electric** | Monitor and measure electricity consumption and | [47], [78], [87] |

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| | | |
|---|---|---|
| **Meters** | energy usage in real time. This category also includes energy production sensors. | |
| **Livestock Monitoring** | Enhance the management and well-being of livestock. This includes Collar sensors, which track the location and movements of animals; Animal Behavior sensors, which monitor patterns such as activity, stress, and health indicators; and Feed Consumption sensors, which measure and optimize the amount of feed consumed by each animal to ensure better nutrition and health outcomes. These sensors collectively support more efficient livestock management and improve animal care. | [25], [58] |
| **Others** | Intelligent Sensors with Cryptographic Module, Modular BlockChain (MBC) Sensing Instrument, SCADA Systems, Ultrasonic (HC-SR04), Potentiometer, Crop Health, and Equipment Status sensors. | [9], [18], [44], [47], [61], [74] |

It becomes evident that sensors like humidity, temperature, and soil moisture play a significant role in monitoring environmental conditions and supporting tasks such as crop health monitoring and irrigation management. Furthermore, the use of GPS and IoT devices highlights the growing trend of integrating these technologies with blockchain for precise location tracking and real-time data management.

### 6.2 Actuators

Several actuators are documented in the literature. For example, Hang et al. [67] use *water pumps* to control the water level in the tanks, adjusting flow rates and durations based on real-time data and predictions. Similarly, *pond heaters* manage water temperature, while *fish feeders* automatically distribute feed to the fish based on predefined conditions. Additionally, *lighting (LEDs)* is controlled to maintain optimal lighting within the tanks.

Jamil et al. [24] utilize various types of actuators, including *heaters, chillers, CO2 generators, humidifiers, fogging systems,* and *ventilators (both natural and forced)*. These actuators are controlled by the output of an optimization module that processes sensor data (such as temperature, humidity, and CO2 levels). Once the optimization algorithm determines the ideal environmental conditions for plant growth, the control module sends commands to the actuators to adjust their settings accordingly.

In the system described by Sakthi & DafniRose [36], *motion control* is part of the broader infrastructure enabling PA. Motion control actuators in this context are used to automate irrigation tasks based on the data collected by sensors in the agricultural environment. Devi et al. [77] describe how various control nodes are integrated into a smart agriculture system that combines IoT and blockchain. These nodes are responsible for managing different environmental factors in agricultural settings. For example, the *Illuminance Control Node* ensures the proper lighting conditions for crops, while the *Wind Speed and Air Control Node* manages airflow and air quality. Similarly, the *CO2 and Pressure Control Node* regulates CO2 levels and atmospheric pressure, which are crucial for plant health.

The *Pollution Control Node* monitors air quality to ensure that pollutants do not affect the environment, and the *Moisture and Water Control Node* tracks soil moisture and regulates irrigation systems. The Smoke and Fire Control Node detects any smoke or fire, automatically triggering alarms or water pumps if needed, providing safety to the crops. Lastly, the *pH Control Node* manages soil or water pH levels, maintaining an optimal environment for plant growth.

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Finally, Enescu et al. [47] utilize *SCADA (Supervisory Control and Data Acquisition) systems* in the context of managing photovoltaic-powered irrigation systems. SCADA is used to monitor and control energy and water consumption in agricultural setups that are powered by renewable energy sources, specifically solar power.

### 6.3 Devices

Every device used in the PA systems that does not fall into the sensor or actuator category is listed in Table 3.

*Table 3. Devices*

| Device | Description | Agriculture task |
|---|---|---|
| **IoT** | Refers to interconnected devices that collect, exchange, and process agricultural data in real-time, enabling smart farming applications and automation. | Monitoring [21], [41] |
| **UAV** | Aerial devices typically carrying various sensors (i.e thermal sensors, RGB cameras, multispectral cameras, hyperspectral cameras, and LIDAR), for information collection, and various other tasks. | Livestock Monitoring and Control [19], Crop Growth Monitoring, Monitoring [37], [62], Insect and Disease Detection [80] |
| **Cameras** | Used in both stationary and mobile setups for visual monitoring, crop health analysis, and pest detection through image processing. | Monitoring [12], [37], [73], [84], Disease Detection [83], Insect Detection [34] |
| **RFID** | A system that uses radio waves for automatic identification and tracking of livestock, equipment, or agricultural products. | Monitoring [12] |
| **Photovoltaic Panels** | Devices that convert sunlight into electricity, providing sustainable energy for powering remote agricultural systems and devices. | Irrigation Control [47] |
| **Solar Energy Charging Piles** | Solar-powered stations used to charge electric agricultural vehicles and equipment, promoting eco-friendly farming. | Water and Energy usage Monitoring [87] |
| **Combines** | Multifunctional machines that perform harvesting, threshing, and cleaning of grains in one operation, streamlining crop collection. | Harvesting [49] |
| **Transporters** | Vehicles or machinery used for moving harvested crops, tools, or other materials across the farm efficiently. | Crop Transport [49] |
| **Tractors** | Versatile vehicles used for towing, plowing, planting, and other heavy-duty farming tasks, often integrated with smart technologies. | Planting [49] |
| **Scout (Field Scouts or Scouting Devices)** | Tools or mobile units used to gather field data, observe crop conditions, and detect issues like pest outbreaks or disease. | Field Exploration [15] |

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| | | |
|---|---|---|
| **Cultivators** | Mechanical devices used to stir and aerate the soil, control weeds, and prepare the ground for planting. | Planting [27] |
| **Seeders** | Machines that plant seeds at the correct depth and spacing, improving planting efficiency and consistency. | Planting [27] |
| **Agricultural Robots** | Automated machines capable of performing tasks like planting, weeding, harvesting, and monitoring with minimal human intervention. | Planting and Crop Monitoring [27] |

## 7. Edge Layer technologies

In this section, we outline in Table 4 the technologies of the Edge layer that serve as the interface between the physical world and the digital infrastructure in system architecture. Positioned close to data sources such as sensors and IoT devices, this layer is responsible for real-time data collection, initial processing, and filtering before transmitting relevant information to higher layers.

*Table 4.Edge Layer technologies*

| Category | Sub Category | Description | Articles |
|---|---|---|---|
| **IoT and hardware** | ESP32 microcontroller | Powerful dual-core microcontroller with integrated Wi-Fi and Bluetooth, ideal for IoT projects | [8], [31] |
| | Arduino Sensor Kit | Includes various plug-and-play sensors compatible with Arduino for learning and prototyping | |
| | ESP8266 NodeMCU | Low-cost Wi-Fi microcontroller board based on the ESP8266, used for simple IoT applications | [23], [57], [71], [81] |
| | Arduino UNO | Popular open-source microcontroller board based on the ATmega328P, widely used in prototyping | [18], [32], [45] |
| | 6V DC Motor | Basic electric motor that operates on 6V DC power, used in small robotics or mechanical projects | [57] |
| | Adjustable DC-DC Power Converter | Converts and regulates voltage between DC power sources, with adjustable output | |
| | MSP430-based Solar Energy Harvesting Tool | Low-power microcontroller platform designed for energy harvesting applications using solar power | |
| | Single-Channel | Electrical switch module | |

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| | | | |
|---|---|---|---|
| | Relay Module | controlled by a microcontroller to switch high-voltage devices | |
| | Microcontrollers (General) | Programmable integrated circuits used for controlling devices and processes | [40] |
| | Arduino (General) | | [86], [64] |
| | ThingsBoard | Open-source IoT platform for device management and data visualization | [12] |
| | Adafruit Python DHT library | Sensor communication and environmental monitoring | [15] |
| | Relay-Aided Wireless Networks | Networks where intermediate devices (relays) help forward data from a source to a destination | [87] |
| | Decode-and-Forward (DF) Relay Strategies | One of the main relaying protocols in cooperative communication | [78], [87] |
| | Raspberry Pi | Small, affordable, single-board computer | [34], [40], [42], [43], [45], [52], [73], [79] |
| **Edge and Fog technologies** | Both | | [36], [39] |
| | Edge Computing | Distributed computing paradigm that brings computation and data storage closer to the sources of data | [10], [26], [40], [48], [55], [58], [60], [68], [75] |
| | Fog Computing | Decentralized computing infrastructure in which data, compute, storage, and applications are distributed in the most logical, efficient place between the data source and the cloud | [11], [12], [17], [18], [37] |
| **Networking and Communications** | LoRaWAN | Low-power, long-range wireless communication protocol | [25], [44], [45], [76], [85] |
| | Nginx and Keepalived Services | Web server and a high-availability solution for load balancing and failover, respectively | [44] |

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| | | | |
|---|---|---|---|
| | Infura API | Service providing access to Ethereum blockchain networks without requiring a full node | [8], [9], [31] |
| | SDN | Enables centralized control of network traffic | [12] |
| | Network Function Virtualization (NFV) | virtualizes network services traditionally run on proprietary hardware | |
| | Open vSwitch | Virtual switch for network automation in virtualized environments | |
| | ONOS | Open-source SDN controller for managing large-scale networks | |
| | Mininet | Network emulator for testing SDN applications | |
| | Hping | Network testing and security auditing | |
| | sFlow | Network traffic sampling and analysis | |
| | MQTT | Lightweight messaging protocol | [9], [25], [42], [48], [63], [69], [70] |
| | RF Transceivers | Transmitting and receiving radio frequency signals | [40] |
| | RFID | Wireless identification and object tracking | [81] |
| | Wi-Fi | Wireless networking | |
| | TCP Sockets | | |
| | 6G | Next-generation wireless communication standard | [72] |
| | CoAPBench | Evaluating the performance of the CoAP protocol | [46] |
| | Pegasus Routing Protocol | Enhances network efficiency and scalability | [55] |

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| | | | |
|---|---|---|---|
| | Smart-M3 | Middleware platform that enables interoperability in smart environments | [15] |
| | D2D Authentication | security mechanism for direct authentication between devices | [26] |
| | D2G communication | Ensures secure interactions between devices and gateways | |
| | Nakagami-m, Rice, and Rayleigh Channels | Statistical models for characterizing wireless communication channels and signal propagation | [78] |
| | IoFT | IoT-based framework for smart agriculture applications | [11] |
| | Internet of Agro-Things (IoAT) | IoT-based architecture optimized for agricultural applications | [48] |
| | Web3.js Library | JavaScript library for interacting with Ethereum smart contracts | [84] |
| | EMQX | Open-source MQTT broker | [42] |
| **WSN** | ILEACH algorithm | Enhanced version of the LEACH protocol | [52], [53] |
| | LEACH (Low-Energy Adaptive Clustering Hierarchy) algorithm | Routing protocol designed specifically for WSNs to reduce energy consumption and extend the network's lifetime | [75] |
| | | | [20], [55], [78] |

## 8. Services Layer Technologies

### 8.1. Blockchain technologies

In the PA domain, blockchain technology is implemented using various platforms to support agricultural tasks and manage data effectively. The primary platforms identified are Ethereum, Hyperledger, IOTA, Gcoin, and EOS, each offering distinct features suited to agricultural needs. Figure 4 shows all the platforms mentioned in literature and Table 5 shows a comparison between them.

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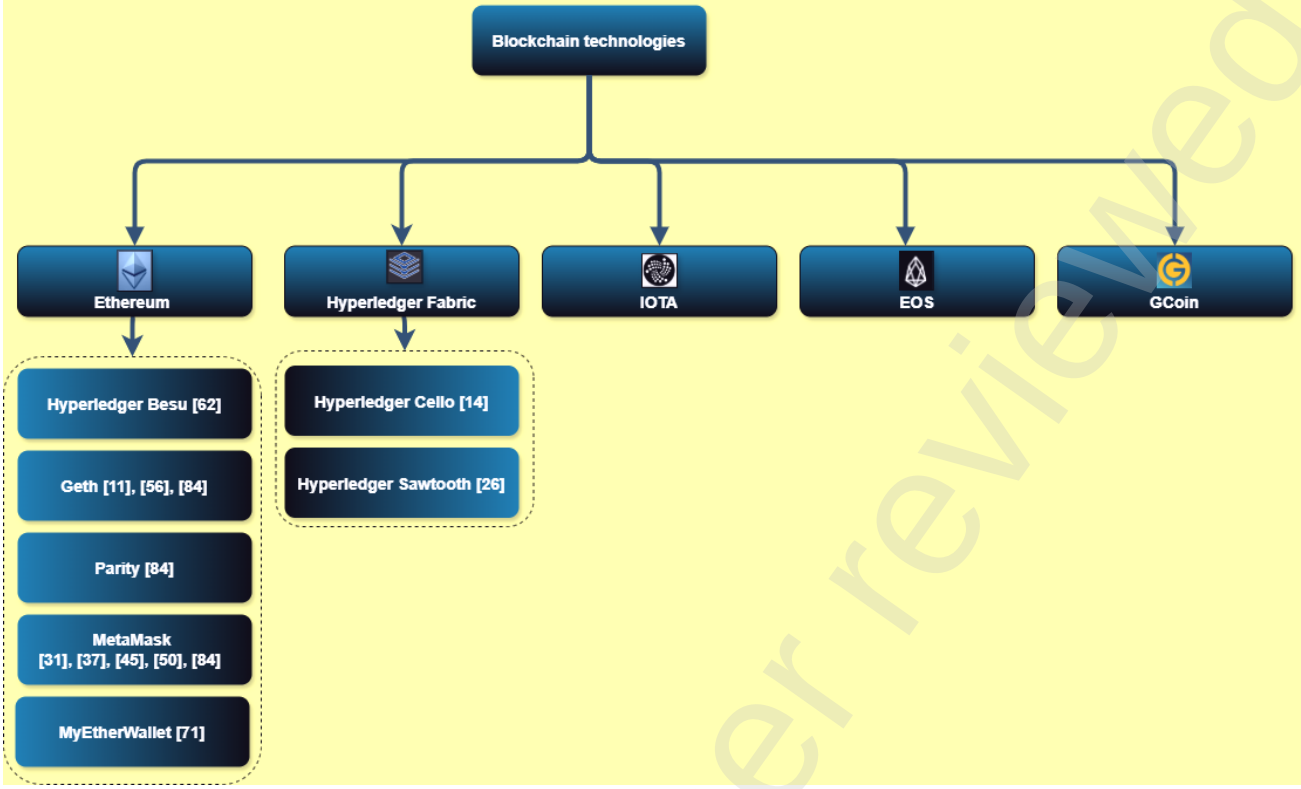


*Figure 4. Blockchain Technologies*

*Table 5.Blockchain platform comparison*

| Criteria | Ethereum | Hyperledger Fabric | IOTA | EOS | GCoin |
|---|---|---|---|---|---|
| Type | Public blockchain | Private/consortium blockchain | Public blockchain | Public blockchain | Private/consortium blockchain |
| Open Source | Yes | Yes | Yes | Yes | No |
| Consensus Mechanism | Proof of Work (PoW), Proof of Stake (PoS), Proof of Authority (PoA) | Practical Byzantine Fault Tolerance (PBFT), Solo, Raft, Proof of Stake (PoS) | Proof of Work (PoW) | Delegated Proof of Stake (DPoS) | Centralized consensus |
| Privacy | Transaction details are public | High privacy with permissioned membership | Transaction details are public | Transaction details are public | High privacy, transactions are confidential |
| Purpose | Smart contracts, transparency, Data integrity, auditability | Permissioned data sharing, Privacy-focused applications | Efficient micro-transactions and scalability, Edge/fog integration | High performance and fast transaction speeds | Data access control |

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| | | | | | |
|---|---|---|---|---|---|
| Use cases | Monitoring, Control (Automation)/ Management, Prediction | Monitoring, Control (Automation)/ Management, Prediction | Monitoring, Control (Automation)/ Management | Monitoring | Monitoring |

Some studies emphasize the type of blockchain used rather than specific platforms. Arsyad et al. [61] propose a Modular Blockchain (MBC)-based farm transaction model to improve data integrity and security. Their system, EBM, integrates sensors, controllers, and NFC-based communication to securely record encrypted transactions in low-connectivity areas.

Wu & Tsai [82] design a private blockchain that protects agricultural data via identity verification, encryption, and dark web masking to block DDoS attacks and unauthorized access. Jadav et al. [72] develop a public blockchain using IPFS and smart contracts to securely store pesticide data and prevent tampering by stakeholders. Luo et al. [27] introduce a hybrid blockchain for cross-domain authentication in smart agriculture IoT, combining private and consortium chains for secure, scalable many-to-many authentication. Vangipuram et al. [48] use BaaS and an IoT Hub to manage weather data, ensuring integrity and accuracy through blockchain-based validation.

The prevalence of Ethereum and Hyperledger is particularly notable, with Ethereum being the most widely used across the reviewed papers. This dominance is likely attributed to Ethereum's ability to support decentralized applications (dApps) and smart contracts, which are crucial for automating various agricultural processes.

### 8.2. Smart Contracts

Smart contracts were identified in the majority of the literature, demonstrating their prominent role in the studied literature. Smart contracts are self-executing programs that run on a blockchain, automatically enforcing terms and conditions of an agreement without intermediaries. They provide transparency, security, and automation, making them particularly useful for applications requiring trustless execution and immutable record-keeping.

Aliyu & Liu's Smart Farm framework [8], uses smart contracts to control IoT device access to resources like water and fertilizers. Another system [29] applies smart contracts in a crop recommendation setup, using real-time sensor data to trigger automated actions like irrigation or crop suggestions, with all actions recorded on the blockchain for transparency. AgriSecure [12] combines blockchain with SDN to secure IoT data sharing and access, using Ethereum smart contracts to validate transactions and maintain a tamper-proof ledger. A similar decentralized system [41] enforces access control by authenticating IoT devices based on roles and attributes via multiple smart contracts.

Several works employ *Remix IDE* to develop and deploy Ethereum smart contracts. Jadav et al. [72] store agricultural data securely, while Bhattacharya & Dey [71] manage IoT sensor data like temperature and soil moisture. Widayat & Köppen's simulated model [84] deploy smart contracts that validate and store encrypted images captured by surveillance cameras, ensuring the authenticity of data in the agricultural supply chain. This research along with [62] include the use of *Truffle framework*, a development framework for smart contracts, allowing for testing, compiling, and deploying the smart contract system.

Dewangan and Vij [58] present a platform that leverages *oracles* to enable communication between IoT devices and the blockchain. These oracles play a crucial role in data validation. When IoT devices gather data, it is initially sent to the oracles for verification. The oracles process the data using the SHA-256 algorithm, generating a hash value. This hash value is then recorded on the blockchain. Once verified, the

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data is transmitted to the blockchain, where it can be utilized to execute smart contracts.

### 8.3. Consensus Mechanisms

Consensus mechanisms are protocols or algorithms employed in blockchain networks to ensure agreement among distributed nodes regarding the network's state, such as validating transactions or making collective decisions. The most commonly discussed mechanisms in the reviewed literature include *Proof of Work (PoW)* [13] [19] [31] [40] [50] [56] [66] [70] and *Practical Byzantine Fault Tolerance (PBFT)* [17] [24] [26] [38] [67]. Other notable mechanisms are *Proof of Authority* (*PoA) (*Clique [62]), Solo*, Raft* [44], *Proof of Stake* [21], Delegated Proof-of-Stake (DPoS) [30], and Proof-of-Performance (PoP) [74]. Additionally, Mehare & Bartere [39] proposed a *new consensus* model that involves a larger number of low-computing-capacity nodes validating new blocks as an alternative approach.

### 8.4. IPFS

IPFS is a decentralized file storage and sharing protocol designed to make the web more distributed. It allows users to store and share files in a way that doesn't rely on central servers, offering a peer-to-peer (P2P) network where files are distributed across multiple nodes. The papers using IPFS are shown in Table 9.

### 8.5. Security services

***Security*** encompasses various techniques designed to enhance the safety and resilience of systems. Noor et al. [41] propose a decentralized framework that integrates *Attribute-Based Access Control (ABAC)* and *Role-Based Access Control (RBAC)* with blockchain technology to secure IoT devices in smart farming. This approach strengthens access management while ensuring a robust and transparent security model.

Sharma & Villanyi [21] leverage *Physical Unclonable Function (PUF)* as a hardware security primitive to authenticate IIoT devices and storage nodes. By integrating PUF with Ethereum's merge-based blockchain technology, their system enables secure, efficient, and sustainable Big Data collection and dissemination in IIoT environments.

Wu & Tsai [82] incorporate *dark web technology* to enhance the privacy and security of a private blockchain-based intelligent agriculture network. In this system, dark web technology functions as a security layer, mitigating cyber threats such as unauthorized access, hacking, data breaches, and DDoS attacks. It ensures that only verified users can interact with the blockchain while concealing network nodes from external threats.

Arshad et al. [14] implement *Private Blockchain Access Control (PBAC)* to strengthen security and privacy in agriculture monitoring systems.

We came across several different ***Encryption Techniques*** referenced in various articles, showcasing a diverse range of cryptographic approaches used in different applications. Each technique, listed in Table 6, serves a unique purpose, from securing communications to enabling efficient authentication and privacy-preserving computations.

*Table 6. Encryption Techniques*

| Encryption Technique | Description | Articles |
|---|---|---|
| RC5 Encryption Algorithm | A symmetric block cipher known for its simplicity, efficiency, and variable-length key support, commonly used in IoT and embedded systems due to its adaptability. | [44] |
| Merkle Tree | A cryptographic structure widely used in blockchain technology for efficient and secure data | [21], [44], [50] |

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| | | |
|---|---|---|
| | verification, ensuring the integrity of transactions and stored information. | |
| Elliptic Curve Digital Signature Algorithm (ECDSA) | A widely used digital signature scheme in blockchain networks for authentication and non-repudiation, providing strong security with shorter key lengths. | [21] |
| Ciphertext-Policy Attribute-Based Encryption (CP-ABE) | An encryption scheme allowing fine-grained access control, where decryption permissions are based on attributes, making it highly useful in privacy-preserving data-sharing systems. | [35] |
| Bilinear Pairings Cryptography | A cryptographic technique used in advanced encryption schemes like identity-based encryption and at-tribute-based encryption for flexible and efficient access control. | [82] |
| Advanced Encryption Standard (AES) | One of the most widely used symmetric encryption tech-niques, offering strong security and efficiency in data encryption for cloud storage, secure communications, and financial transactions. | [68], [82], [86] |
| Masked Authenticated Messaging (MAM) | A protocol enabling secure, encrypted messaging in IoT networks, often used in decentralized environments for maintaining data privacy and integrity. | [40] |
| Public Key Infrastructure (PKI) | A framework managing encryption keys and digital certificates for secure communications, essential in web security, electronic signatures, and identity verification. | [67] |
| BLS Short Signature | A cryptographic signature scheme that allows short signatures with batch verification, improving efficiency in blockchain applications and secure authentication. | [17] |
| Paillier Encryption | A homomorphic encryption scheme allowing computations on encrypted data without decryption, widely used in privacy-preserving computations and secure data analysis. | [11], [17] |
| DVAC (Designated-Verifier Anonymous Credential) | A cryptographic method that ensures user anonymity while allowing only designated verifiers to validate credentials, used in privacy-preserving authentication. | [18] |
| Data Encryption Standard (DES) | An older symmetric encryption algorithm, used for securing digital communications, largely replaced by AES due to security concerns. | [33] |
| Multi-Image Encryption (MIE) Algorithms | Techniques designed for encrypting multiple images simultaneously, used in secure image | [84] |

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| | | |
|---|---|---|
| | transmission, medical imaging, and multimedia security. | |
| Elliptic Curve Cryptography (ECC) | A public-key encryption system providing strong security with smaller key sizes, commonly used in secure web communications, digital signatures, and blockchain. | [16], [38], [62] [26] [27] |
| CertificateLess Cryptography (CLC) | A cryptographic system that eliminates the need for a certificate authority while maintaining security, reducing overhead in key management. | [27] |
| Groupable Batch Verification (GBV) Algorithm | An authentication mechanism that allows the verification of multiple signatures in a single process, improving efficiency in blockchain and network security. | [27] |
| RSA Encryption | Widely used asymmetric encryption algorithm, known for its strong security in key exchange, secure communications, and digital signatures. | [86] |
| Homomorphic Encryption and Secret Sharing | A combination of cryptographic methods enabling computations on encrypted data and secure data division among multiple parties for privacy-preserving applications. | [80] |

The diverse range of encryption techniques referenced in the articles highlights the varying security needs across different domains. Traditional symmetric algorithms like AES, RC5, and DES focus on speed and efficiency, while ECC, RSA, and ECDSA provide strong security for authentication and digital signatures. Advanced cryptographic techniques such as homomorphic encryption, attribute-based encryption (CP-ABE), and bilinear pairings address complex privacy and access control challenges.

The available literature discusses the use of several ***hash functions***, each serving specific security and efficiency purposes. *SHA-256* is the most commonly used, appearing in multiple articles [12] [14] [17] [26] [50] [62] [68] [71]. It is a widely adopted cryptographic hash function known for its strong security properties and is frequently used in blockchain technology and secure data hashing due to its resistance to collision and preimage attacks.

*BLAKE3* appears in Sharma & Villanyi's [21] study and is a modern cryptographic hash function designed for high speed and efficiency. It provides strong security while outperforming traditional hash functions like SHA-256 in terms of performance.

*HMAC (Hash-based Message Authentication Code)* is used by Wu & Tsai [82], emphasizing its role in secure communications and authentication. By combining a cryptographic hash function with a secret key, HMAC ensures data integrity and authenticity.

*Hash Merging* employed by Zhao et al. [44], suggesting an effort to enhance security or efficiency by leveraging multiple hash functions together. This approach can help reduce collisions and improve hashing performance.

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Overall, SHA-256 dominates the selection of hash functions in these articles, while BLAKE3 and HMAC reflect the consideration of alternative methods for improved security and efficiency.

### 8.6. Cloud services

Many of the selected papers employ ***Cloud*** technology, primarily for data storage and management. Hossain et al. [57] utilize the *Firebase Cloud Database* for real-time data access, remote monitoring, automation, and secure data handling. Meanwhile, Arsyad et al. [61] leverage *Google Cloud* to store and analyze transaction data from cocoa processing, while Bhattacharya & Dey [71] employ Google Cloud for data storage, smart contract execution, and real-time monitoring. Frikha et al. [73] use Google Colab to write and execute Python code in a Jupyter notebook environment.

*AWS (Amazon Web Services) Cloud* is used by some systems for storing, processing, and analyzing sensor data. Specifically, *AWS IoT core* is employed to gather messages from diverse IoT devices and *AWS Lambda* to perform the cryptography and save the data to the blockchain [8] [9] [31]. Lamtzidis et al. [40] utilize *AWS EC2* as a scalable cloud computing resource to support IoT-based smart farming applications. It provides secure and resizable compute capacity for processing sensor data from IoT devices in agricultural fields.

Sangeetha et al. [45] utilize the Azure SQL Database to store real-time sensor data, while Palacios et al. [65] develop a system that uses Azure Functions to create tasks, assign users, and streamline farm management operations. In addition, Twilio is integrated to enable real-time SMS communication, enhancing the farm management system's messaging capabilities. Furthermore, Vangipuram et al. [48] leverage Azure IoT Hub to enable bi-directional communication between IoT sensor devices, which collect weather data.

Sakthi & DafniRose [36] employ *ThingSpeak* as the cloud platform for aggregating, visualizing, and analyzing live data streams in their blockchain-enabled smart agricultural knowledge discovery system.

### 8.7. Testing tools

Table 7 presents the most commonly used blockchain testnets and security/performance testing frameworks identified in the literature.

*Table 7. Testing tools*

| Tool | Description | Articles |
|---|---|---|
| *Ethereum Rinkeby* | Test networks for evaluating smart contract mechanisms | [8], [9], [31] |
| *Sepolia* | | [29] |
| *Ropsten* | | [50] |
| *Goerli* | | [68] |
| *Apache AB* tool | A benchmarking tool used for performance testing of web applications and APIs. It can simulate concurrent requests and measure response times, useful in blockchain-integrated web interfaces. | [44] |
| *Automated Validation of Internet Security Protocols and Applications (AVISPA)* | A formal security analysis tool that validates cryptographic protocols. It helps ensure the correctness of authentication and | [26], [38] |

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| | | |
|---|---|---|
| | communication security in blockchain-based systems. | |
| *Hyperledger Caliper* | A blockchain benchmarking tool that allows developers to test the performance of blockchain networks, including Hyperledger Fabric, Ethereum, and others, by generating reports on TPS, latency, and resource usage. | [24], [43], [62], [67] |

### 8.8. Database services

Several studies emphasize the integration of databases within their systems. In Pullo et al.'s work [42], MongoDB is seamlessly combined with Raspberry Pi devices, MQTT protocols, and cloud-based dashboards to support efficient data monitoring. By utilizing MongoDB Atlas, the researchers ensure secure storage, scalability, and structured data processing. This database facilitates the visualization of critical agricultural metrics, including temperature, humidity, and sensor activity over time. Similarly, in Lamtzidis et al.'s study [40], MongoDB is integrated with backend technologies like Node.js to enable dynamic CRUD (Create, Read, Update, Delete) operations, allowing users to manage data effortlessly.

Lin et al. [66] describe an ICT e-agriculture system where an SQL database is used for querying, uploading, and downloading data, providing a user-friendly interface for record management applications. In this system, blockchain ensures immutability and decentralization, while the SQL database handles structured storage and off-chain data management.

Hang et al. [67] employ CouchDB as a distributed database architecture within their blockchain-based fish farm platform. CouchDB is deployed on each blockchain peer to enable efficient file storage while minimizing duplication across the blockchain filesystem. This setup allows for off-chain storage, ensuring that large amounts of agriculture data can be effectively managed without overloading the blockchain itself.

Quafiq et al. [22] describe *ODBC (Open Database Connectivity)* and *JDBC (Java Database Connectivity)* drivers as part of the connectivity patterns used to link various data sources to the data lake. These drivers facilitate extracting data from RDBMS (Relational Database Management Systems) and other structured data sources into the Big Data architecture.

Frikha et al. [73] utilize a *MySQL* database in the implementation of a smart greenhouse system, where it records data from sensors measuring parameters like temperature, humidity, soil moisture, and other environmental conditions.

## 9. Application Layer Technologies

### 9.1. Web and Application Development

In the domain of **Web & Application Development**, we incorporated several technologies that are presented in Table 8.

*Table 8. Web & Application Development technologies*

| Technology | Description | Articles |
|---|---|---|
| Web Application Graphical User Interfaces (GUIs) | They enhance user experience by enabling easy navigation and data input/output. | [8] |
| ReactJS | A popular JavaScript library for | [50] |

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| | | |
|---|---|---|
| | building fast, dynamic, and reusable user interfaces, especially for single-page applications. | |
| ASP.NET Core Framework | A cross-platform, high-performance framework by Microsoft for building modern, cloud-based web applications and APIs. | [45] |
| Entity Framework 5 | An Object-Relational Mapping (ORM) tool for .NET that allows developers to interact with databases using .NET objects, simplifying data access code. | [45] |
| Flask | A lightweight, Python-based web framework known for its simplicity and flexibility in building web applications and REST APIs. | [22] |
| RESTful APIs | They follow the REST architecture and enable systems to communicate over HTTP using standard methods like GET, POST, PUT, and DELETE, making integration easy and scalable. | [22], [23], [67], [70], [72] |
| Visual Studio Code | A free, open-source code editor by Microsoft with support for debugging, syntax highlighting, Git, and extensions for many programming languages. | [68] |
| Node.js | A runtime environment that allows JavaScript to be used for server-side development, known for its event-driven, non-blocking I/O model and scalability. | [81] |
| MIT App Inventor | A block-based programming environment that enables users, especially beginners, to create fully functional Android apps without writing traditional code. | [57] |
| Docker | A platform that uses containerization to package applications and their dependencies together, ensuring they run reliably across different environments. | [12], [22], [44], [67], [70] |

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| | | |
|---|---|---|
| Apache web server | A widely used open-source web server software that delivers web content to users' browsers and supports various modules for added functionality and performance. | [73] |

### 9.2. AI applications

AI has shown promising results in agriculture, including its integration with UAVs for aerial monitoring tasks. For example, in saffron cultivation, UAV imagery combined with machine learning models has been successfully used for detecting flowers, weeds, and animal intrusions [88]. Many techniques and technologies are incorporated in the AI category. ML [10] [29] [57] [36] [67] [77] [55] [22] [85] [58] [34] [59] [28] [63] was the most commonly used AI technology, followed by DL [20] [83] [18] [73] [34].

Jadav et al. [72] apply Support Vector Machine (SVM), Decision Tree, Random Forest, and Perceptron classifiers to predict pesticide thresholds in crops, aiming to minimize health risks. The BAFL-SVM framework [80] integrates SVM, Federated Learning, and Image Preprocessing Techniques for smart agriculture, specifically for rice pest and disease detection.

TensorFlow, along with Keras, is used in various applications: Devi et al. [83] integrate it with ResNet for plant leaf disease detection, and Frikha et al. [73] utilize it for monitoring plant growth and environmental conditions. Sonavane et al. [74] propose a bioinspired Q-learning model to optimize Quality of Service (QoS) and security in AIoT deployments. Their approach integrates Q-Learning for Miner Selection (where reinforcement learning is used to dynamically select miners based on performance and reliability), Mayfly Optimization (MO) for Shard Configuration (which balances load and enhances transaction efficiency by mimicking the swarm behavior of mayflies), and Bacterial Foraging Optimization (BFO) for Sidechain Optimization (which improves blockchain scalability and security by simulating bacteria’s foraging strategies to find the most optimal sidechain structure).

Mahalingam & Sharma's system [16] uses Recurrent Neural Networks (RNNs) in their Recurrent Neural Elliptical Curve Blockchain (RNECB) framework to process time-series agricultural data (e.g., temperature, humidity) for real-time monitoring and anomaly detection. Jegadeesan et al. [18] use the Contractive Sparse Autoencoder (CSAE) and SoftMax Classifier as part of their DL architecture to detect intrusions in an IoT-enabled environment for smart farming.

The system developed by Ting et al. [32] employs Intelligent Fuzzy Logic as a decision support system for managing the watering process in an IoT-based smart climate agriculture system. The system described by Ahmed et al. [75] utilizes Fuzzy Matrix in the data aggregation process for smart agriculture. Specifically, the fuzzy similarity matrix is employed to group sensor nodes based on data similarity, which enables efficient data transmission while reducing redundancy and energy consumption. Dewangan & Vij [58] use a Fuzzy Logic Expert System (FLES) to support decision-making in livestock management by handling complex and uncertain conditions.

Rao et al. [54] employ Distributed Fuzzy Cognitive Maps (DFCM) and Earthworm-based Deer Hunting Optimization Algorithm (EW-DHOA) in the context of WSNs for smart agriculture to enhance efficiency, security, and resilience. Similarly, Rajeshkumar & Sureshkumar [55] use the Ant Lion Optimization (ALO) Algorithm in their research to enhance energy-efficient routing in WSNs within an IoT-based agricultural framework. Their approach, termed HB-ALO, integrates blockchain with ALO for optimized data transfer and network efficiency. Lastly, the Kalman Filter Algorithm is employed by Hang et al. [67] for predicting greenhouse sensor data, such as temperature, humidity, and CO2 levels, and by Jamil et al. [24] to remove

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errors from water level sensor readings.

### 9.3. Simulation and OS Software

The ***Simulation Software & OS*** category includes a diverse range of tools that support simulation, electronics design, robotics, and blockchain environments. It features *a Linux-based OS* [81], a versatile open-source operating system widely used in embedded systems and simulation platforms. *Fritzing*, an open-source electronics design automation (EDA) software [57], provides users with a platform for creating circuit diagrams, PCB layouts, and prototypes. The category also includes *RTOS (Real-Time Operating System)* [31], which is essential for real-time, deterministic processing in embedded and control systems. *MATLAB* [51] is a high-level programming environment often utilized for numerical computation, algorithm development, and data visualization in engineering and scientific applications. *Gazebo*, a robust simulation platform for robotics [49] [15], enables realistic simulation of robotic systems in various environments. *Proteus* [18] is a simulation and design software that combines circuit simulation, PCB design, and microcontroller simulation into one comprehensive tool. Finally, *Ganache* [43], [45], [50], [71], [84] is the most widely used blockchain simulator, providing developers with an environment to test and develop Ethereum-based decentralized applications.

### 9.4. Management Tools

We identified three technologies in the literature, which we have categorized as ***Management Tools***. Palacios et al. [65] employ *ClickUp* as part of their Farm Management System to automate task creation, user assignment, field geolocation, and chat mirroring. They integrate ClickUp automations, webhooks, chat views, users, and custom fields to provide real-time monitoring and task management. Chang et al. [46] utilize *2D modeling* in conjunction with IoT and blockchain to enhance control and management in smart water systems for agriculture. By integrating real-time data collection from IoT sensors, they create a digital representation of water distribution and flow.

Finally, Ahmed et al. [75] use *sleep scheduling* as a way to optimize data aggregation in smart agriculture. By organizing sensors into clusters, they assign each cluster a leader responsible for managing data collection and transmission. To conserve energy and reduce redundancy, a fuzzy similarity matrix is applied to identify nodes that provide overlapping data. These redundant nodes are placed in sleep mode while others remain active, ensuring efficient monitoring without unnecessary power consumption.

### 9.5. Data technologies & Techniques

Lastly, we categorized several technologies under the ***Data Technologies & Techniques*** umbrella. Chang et al. [46] use *Complex Event Processing (CEP)* technology in their smart water management system to analyze real-time data collected from IoT sensors. By processing recurring patterns, CEP helps detect water loss and inefficiencies in the system. When anomalies are identified, the system triggers alarms, allowing water management agents to execute corrective actions. This approach integrates CEP with IoT-based remote monitoring, ensuring dynamic decision-making in response to changes in water supply conditions.

In their study, Quafiq et al. [22] employ *Big Data* technologies to enhance smart farming and integrate blockchain for secure data processing. Their architecture incorporates *Hadoop* for large-scale data storage and processing, *Spark* for efficient in-memory computation, and data ingestion tools like *Sqoop, Flume, Kafka, and NiFi* to manage the flow of agricultural data from various sources, including IoT devices and sensors. These technologies enable real-time and batch data processing, ensuring data integrity, scalability, and interoperability within a distributed framework for smart farming analytics. Similarly, Nguyen et al. [85]utilize *Big Data* in the context of climate-smart and sustainable agriculture, specifically focusing on the Vietnamese coffee supply chain. Their study highlights the role of Big Data in integrating IoT, AI, and

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blockchain technologies for data-driven decision-making in coffee farming.

Finally, Sonavane et al. [74] use *Sharding Techniques* to optimize Quality of Service (QoS) and security in AIoT deployments. Sharding is introduced to enhance scalability, reduce latency, and optimize resource utilization in AIoT networks. The existing blockchain-based security models are often complex, energy-intensive, and have high latency, making them impractical for large-scale deployments.

Table 9 presents all the technologies found in literature with their references.

*Table 9. Technologies*

| Technologies | Articles | Frequency |
|---|---|---|
| Smart Contracts | [8], [9], [12], [15], [16], [21], [23], [24], [25], [26], [27], [28], [29], [30], [31], [34], [35], [36], [37], [38], [41], [43], [45], [46], [47], [48], [49], [50], [51], [54], [56], [58], [59], [60], [61], [62], [63], [64], [65], [67], [68], [69], [70], [71], [72], [73], [75], [76], [77], [80], [81], [82], [84] | 67.5% |
| IoT and Hardware Integration | [8], [12], [18], [23], [31], [32], [34], [40], [42], [43], [45], [52], [57], [64], [71], [73], [78], [79], [81], [86], [87] | 26.3% |
| Web & Application Development | [8], [12], [22], [23], [44], [45], [50], [57], [67], [68], [70], [73], [81] | 16.3% |
| AI | [10], [16], [18], [20], [22], [24], [28], [29], [32], [34], [36], [54], [55], [57], [58], [59], [63], [67], [72], [73], [74], [75], [77], [80], [83], [85] | 33.8% |
| Edge/ Fog Computing | [10], [11], [12], [17], [18], [26], [32], [36], [37], [39], [40], [48], [55], [58], [60], [68], [75] | 21.3% |
| Cloud | [8], [9], [18], [21], [26], [29], [31], [32], [34], [36], [37], [40], [42], [44], [45], [48], [52], [57], [59], [60], [61], [62], [65], [71], [73], [75] | 33.8% |
| Testing Tools | [8], [9], [24], [26], [29], [31], [38], [43], [44], [50], [62], [67], [68] | 17.5% |
| Consensus Mechanisms | [13], [17], [19], [21], [22], [24], [26], [31], [38], [40], [44], [50], [56], [62], [66], [67], [70], [74] | 23.8% |
| IPFS | [19], [30], [35], [50], [60], [62], [68], [69], [72] | 12.5% |
| Simulation Software & OS | [15], [18], [31], [43], [45], [49], [50], [51], [57], [71], [81], [84] | 15.0% |
| Networking & Communication | [8], [9], [11], [12], [15], [25], [26], [31], [40], [42], [44], [45], [46], [48], [55], [63], [69], [70], [72], [76], [78], [81], [84], [85] | 30.0% |
| Encryption Techniques | [11], [16], [17], [18], [21], [26], [27], [33], [35], [38], [40], [44], [50], [62], [67], [68], [80], [82], [84], [86] | 25.0% |
| Hash functions | [12], [14], [17], [21], [26], [27], [44], [50], [62], [68], [71], [82] | 15.0% |
| Security | [14], [21], [41], [82] | 5.0% |
| UAV-Robot Technologies | [15], [19], [37], [49], [56], [60], [62], [80] | 10% |
| Wallet services | [31], [37], [45], [50], [71], [84] | 7.5% |
| Blockchain types | [27], [48], [61], [72], [82] | 6.3% |

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| | | |
|---|---|---|
| Management Tools | [46], [65], [75] | 3.8% |
| Databases | [22], [40], [42], [66], [67], [73] | 7.5% |
| WSN | [20], [52], [53], [55], [75], [78] | 7.5% |
| Data Technologies & Techniques | [22], [46], [74], [85] | 5.0% |

## 10. Blockchain technology evaluation methods

This section discusses the evaluation methods used in the reviewed studies and the metrics employed. In terms of evaluation type, the majority of studies relied on system simulations. Sixteen studies assessed their systems in real-world scenarios, while only six utilized both evaluation types.

The metrics identified in the literature are listed in Table 10, along with the corresponding articles that mention them.

*Table 10. Evaluation Metrics*

| Evaluation Metrics | Articles | Frequency |
|---|---|---|
| Throughput | [8], [9], [12], [13], [24], [27], [30], [31], [40], [43], [44], [51], [54], [56], [62], [67], [74], [77], [78], [87] | 25.0% |
| Execution Time | [8], [16], [33], [40], [46], [56], [57], [58], [60], [65], [82] | 13.8% |
| Network Metrics | [10], [11], [12], [13], [21], [23], [24], [26], [27], [31], [34], [36], [38], [42], [43], [51], [52], [53], [54], [55], [60], [62], [64], [67], [69], [71], [72], [74], [75], [81] | 37.5% |
| Accuracy | [16], [20], [29], [32], [34], [48], [57], [58], [59], [72], [73], [79], [80], [83], [86] | 20.0% |
| Costs | [26], [27], [29], [30], [36], [38], [39], [41], [43], [47], [48], [50], [58], [75], [76] | 18.8% |
| AI Metrics | [20], [34], [54], [55], [57], [59], [60], [83] | 11.3% |
| Transaction Upload Time | [13], [26], [34], [50], [76] | 7.5% |
| Storage and Data Management Metrics | [19], [21], [35], [44], [48], [57], [58], [63] | 11.3% |
| Resource Utilization | [10], [11], [13], [18], [21], [23], [24], [26], [27], [30], [34], [38], [39], [40], [41], [42], [47], [51], [54], [55], [71], [74], [75], [76] | 30.0% |
| Environmental Metrics | [23], [42], [46] | 3.8% |
| Scalability | [13], [15], [22], [36], [43], [67], [72] | 8.8% |
| Security | [11], [14], [16], [17], [18], [22], [26], [27], [28], [30], [35], [36], [38], [44], [60], [61], [64], [66], [68], [69], [70], [71], [75], [79], [80], [86] | 33.8% |
| Transaction Success Rate | [56], [62], [78], [84], [85], [87] | 7.5% |
| Performance & | [9], [10], [11], [13], [16], [24], [25], [28], [29], [35], [36], [42], | 35.0% |

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| | | |
|---|---|---|
| Efficiency | [47], [49], [50], [52], [53], [58], [64], [68], [70], [73], [75], [76], [78], [79], [80] | |
| Usability | [47], [66], [67] | 3.8% |
| Functionality | [19], [26], [38], [57], [69] | 6.3% |

Table 10 presents a comprehensive set of evaluation metrics used to assess blockchain applications in PA, with varying frequencies of appearance across the literature. Some metrics, such as network performance and security, are widely studied, while others, like usability and environmental impact, appear far less frequently.

*Network-related metrics* are the most frequently analyzed, covering factors such as latency, bandwidth usage, and packet loss. Given that blockchain-integrated PA relies on IoT networks, maintaining stable and low-latency communication is essential. *Performance and efficiency*, is another dominant category, reflecting efforts to optimize blockchain frameworks for agricultural applications while minimizing resource consumption.

*Security* is widely discussed, emphasizing blockchain's role in protecting agricultural data, preventing cyber threats, and ensuring data integrity. *Resource utilization*, evaluates computational efficiency and energy consumption, particularly relevant for battery-powered IoT sensors in smart farming. *Throughput* is crucial for assessing how well blockchain systems can handle real-time agricultural data exchanges, ensuring smooth transaction processing.

*Execution time* assesses the speed of smart contract execution and data transactions. This is particularly relevant in scenarios requiring rapid decision-making, such as automated irrigation or pest control systems. *Cost analysis*, focuses on transaction fees, computational expenses, and infrastructure costs, which can impact blockchain's feasibility for small and medium-sized farms.

*Accuracy* is mainly used in AI-integrated blockchain applications to evaluate the reliability of ML models and blockchain-verified data. *AI-related metrics* further assess the performance of AI models used in predictive analytics and decision support within blockchain-based PA.

*Scalability*, addresses blockchain's ability to handle increasing data loads, particularly in large-scale farms with thousands of IoT nodes. *Storage and data management metrics*, evaluate blockchain's efficiency in handling large volumes of sensor-generated agricultural data.

*Transaction upload time* is critical in ensuring timely data recording, while *transaction success rate*, measures the reliability of blockchain networks in validating and storing agricultural data. *Functionality*, evaluates whether blockchain applications meet specific agricultural needs, such as traceability and supply chain transparency.

*Usability*, one of the least explored areas indicating a research gap in making blockchain solutions accessible to farmers and agricultural stakeholders. Similarly, *environmental impact* suggests the need for more research into blockchain's sustainability, particularly in reducing energy consumption through lightweight consensus mechanisms.

In summary, the frequency of appearance of these metrics highlights a strong emphasis on network performance, security, and efficiency, while usability, functionality, and environmental impact remain underexplored. Future research should aim to balance efficiency, scalability, and economic viability while ensuring that blockchain solutions are accessible and practical for PA.

## 11. Discussion

This review has systematically examined 80 peer-reviewed studies to explore the integration of

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blockchain technology in PA. Our findings reveal a dynamic and evolving research landscape, where blockchain is not only used to ensure data integrity and security but also to support automation, trust, and transparency in farming operations. The discussion is structured around three key themes: synthesis of the findings, challenges identified, and opportunities for future work.

### 11.1. Synthesis of Findings

The reviewed studies highlight that blockchain is most commonly deployed in monitoring, control, and prediction tasks within PA. Monitoring applications dominate the field, with use cases including environmental monitoring, crop status tracking, and resource usage (e.g., water and energy). Control-focused systems, often integrated with IoT and smart contracts, automate irrigation, fertilizer application, and task scheduling. Predictive applications, while less common, demonstrate the value of blockchain in disease and pest forecasting, crop prediction, and decision support.

Technologically, Ethereum, Hyperledger, and IOTA are the most utilized blockchain platforms, each selected based on trade-offs between scalability, energy efficiency, and ease of integration with IoT or edge computing. The analysis of architectural components revealed a consistent four-layer architecture (physical, edge, service, and application) that underpins most blockchain-PA applications, underscoring the importance of interoperability across devices, networks, and data layers.

### 11.2. Challenges and Limitations

Through our review of 80 studies, we identified a comprehensive range of challenges that hinder the deployment of blockchain technologies in PA. These challenges span technical, operational, and regulatory domains and are organized into thematic categories for clarity.

Table 11 presents all of the issues documented in the literature, as well as the potential solutions.

*Table 11. Challenges and solutions*

| Challenges | Proposed Solutions |
|---|---|
| **Scalability** [8], [21], [28], [30], [40], [41], [43], [44], [47], [54], [59], [66], [67], [75] | Sidechain to manage transactions that are not important to the main blockchain [8] |
| | Proof-of-stake consensus mechanism rather than energy-intensive proof-of-work and a consortium blockchain [21] |
| | Three-layer architecture in which each layer can accommodate several devices, facilitating system scalability [28] |
| | EOS blockchain [30] |
| | M.E.A.N. (MongoDB, Express, Angular, Node) stack, relatively cheap specialized PoW core for IoT hardware [40] |
| | Combining ABAC and RBAC access control models [41] |
| **Energy constraints** | |
| Ensuring a continuous energy supply for IoT devices [57] | Energy-efficient sensing devices [76] |
| maintaining low energy consumption [21] | Blockchain for its decentralization [43] |
| limited battery life of sensors [52] | Lightweight consensus protocols [74] |

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| | |
|---|---|
| high energy consumption [24], [51], [55], [64], [16], [39], [77] | Energy-aware routing algorithms [54], [75] |
| | Energy-efficient communication technologies [86] |
| **High Resource Consumption** | |
| high computational costs due to cryptographic operations and smart contract execution [11], [17], [23], [24], [26], [27], [38], [39], [44], [69], [72] | Batch writing at the gateway along with a "one transaction, one block" block generation strategy, which enhances the rate of block creation [44] |
| substantial communication overhead between distributed devices [17], [26], [27], [38], [80] | Reducing the number of cryptographic operations required [11] |
| storage requirements for maintaining immutable records [68], [69] | Implementing the system on Cardano or Solana blockchains, to lower transaction fees [9] |
| gas fees and transaction costs [9], [25], [30], [31], [34], [69], [73] | Utilizing the IOTA Tangle along with off-chain storage [13] |
| limited computational and memory capacity of low-cost IoT devices (e.g., Raspberry Pi, ESP32) [13], [28], [33], [34], [64], [77] | Emulating the Simple Network Management Protocol (SNMP) using a REST-based architecture and develop a lightweight micro-operating system for ESP8266 microcontrollers [23] |
| **Data management** | |
| handling massive amounts of big data generated by IoT devices [21], [25], [57], [68], [86] | Combining cloud technologies, which have storage and cost advantages, with blockchain to reduce security risks [21], [57], [84] |
| overcoming storage media constraints [84] | Off-chain storage (MongoDB or Cassandra, IPFS [68]) with a Merkle tree where only the root is stored [25] |
| IPFS effectiveness relies on the number of online peers [19] | Modifying verification batch sizes [27] and leveraging smart contracts and platforms like Smart-M3 to manage distributed data effectively [15] |
| Data coming from heterogeneous sources [22] | |
| **Adoption and integration barriers** | |
| Integration with existing agricultural systems is often | Smart Systems Oriented Big Data Architecture (SSOBDA) that incorporates blockchain using the pBFT consensus mechanism [22] |

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| | |
|---|---|
| hindered by compatibility issues with legacy infrastructures and diverse IoT protocols [22], [47], [51], [52], [67], [76], [79] | Forming farmer associations and implementing an irrigation system that uses photovoltaic energy and blockchain [47] |
| | Leveraging blockchain to mitigate IoT limitations [51] |
| | Customizing blockchain implementations to function alongside existing legacy infrastructure without requiring full system replacement [67] |
| | Fog computing [52], or constrained sensing devices in combination with a permissionless blockchain [76] |
| Adaptability of blockchain systems to various livestock environments [58] | |
| Interoperability [15], [69], [81], [86] | Use of ontologies [15], Ethereum's open-source nature [69], a three-layered IoT architecture combining private and public blockchain [81], and a client-server architecture that promotes interoperability through a common communication interface [86] |
| Requires advanced technical expertise, many farmers have limited digital literacy and access to adequate digital tools and connectivity [47], [75], [85] | Organizing farmer training programs, creating more user-friendly interfaces, providing ongoing technical support, and effectively communicating the benefits of implementing blockchain technology [75] |
| **Complexity** [16], [18], [59], [66], [69], [70], [74] | |
| User interface complexity [67] | More user-friendly application that will visualize blockchain data and enable for real-time control of many aspects [67] |
| Complex channel conditions require the use of sophisticated channel models like Nakagami-m, Rice, and Rayleigh [78], [87] | Use of DF relay model, which helps overcome poor transmission caused by complex channel conditions, along with a relay node selection scheme based on maximizing SIR/SNR [78] |
| | Modeling the wireless channels using the α-μ distribution, a more generalized and flexible channel model compared to the commonly used Rayleigh, Rice, and Nakagami-m models [87] |
| **Network and communication issues** [13] | |
| Lack of sufficient network infrastructure [67] | Flexible BaaS platforms that abstract away the complexity of managing the underlying infrastructure [67] |
| Network connectivity-stability [10], [20], [27], [55], [57], [75], [87] | Signal regenerators to mitigate transmission degradation, often in conjunction with DL methods for real-time attack classification and filtering [20] |
| | ALO algorithm has been explored for achieving stable routing [55] |
| Network lifetime [51] | Smart contracts to prolong network lifetime and enhance operational reliability [51] |

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| | |
|---|---|
| Performance bottlenecks and reduced throughput [10], [84] | Integration of ML techniques to optimize routing and network connectivity [10] |
| Network latency [8], [9], [34], [42], [43], [55], [57], [69], [74], [81] | Lightweight communication protocols [13] |
| Limited bandwidth [20], [57], [81] | combining scalable authentication protocols with adaptive batch processing [27] |
| **Cost** | |
| Infrastructure usage and maintenance [13], [16], [43], [58], [64], [85] | Mathematical model to estimate the infrastructure cost of a blockchain-based crop monitoring system [43] |
| | Improved decision-making and reduced waste enabled by these technologies can offset initial investment costs over time [16] |
| **Operational and technical challenges** | |
| Sensors in controlled environments may produce different results than those in open-field conditions [9] | Systems should also be tested under real-world open-field conditions [9] |
| Adapting predictive models from controlled simulations to actual farm environments [15] | System that combines simulation realism, context-awareness through ontologies, and reliable execution tracking with blockchain [15] |
| Inconsistencies in data formats and sensor readings [22], [70], [80] | Using ETL processes and Big Data integration at the data pipeline and architecture level [22] |
| | Data validation and schema enforcement [70] |
| | Federated learning along with secure preprocessing [80] |
| Handling the volume of data from different sources [23] | Implementing sophisticated systems that only relay data from sensors when there is a major change in conditions [23] |
| Sensor sampling and dynamic responses [42] | Integrating IoT sensors with IOTA's Tangle, which is designed specifically to handle high-frequency, real-time data from IoT devices [42] |
| Using Z-score [56] | Prediction algorithms that can offer more accurate and timely insights [56] |
| bugs and limitations of the ClickUp REST API [65] | Integrating Azure serverless functions, webhooks, and custom parsing logic to handle task creation and management externally [65] |
| **System reliability** | |
| Vulnerability of sensor devices [12] [84] | Fog-computing-based security architecture with blockchain and SDN integration [12] |
| | Loging sensor and surveillance data immutably in a blockchain using Multi-Image Encryption (MIE) techniques [84] |
| Issues with UAVs such as premature descent and loss of stability during landing [19] | a) incorporating real-time monitoring and control via blockchain and IPFS, alongside hardware/software requirement planning; b) designing UAVs with custom hardware tailored for specific agricultural terrain [19] |

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As shown in Table 11, several barriers that hinder the deployment of blockchain in PA, including high cost and lack of knowledge, are also reflected in our previous study about smart farming adoption in Europe [89].

### 11.3. Future Research Directions

Although blockchain has great promise for PA, there are some research gaps that require further examination. Major future directions include:

- **Lightweight Blockchain Architectures:** Adopting blockchain in resource-constrained situations like rural farms necessitates efficient, lightweight consensus systems. Energy-efficient architectures and innovative consensus techniques (such as DAGs, sharding, and PoS variations) optimized for agricultural IoT should be investigated.
- **Interoperability and Standardization:** With the proliferation of blockchain solutions, it's crucial to ensure compatibility across platforms, IoT protocols, and older agricultural systems. Future research should focus on developing middleware, APIs, and ontologies that allow for smooth integration of disparate systems.
- **Scalability and Real-Time Performance:** Blockchain solutions require scalability and real-time performance to handle big agricultural datasets without latency issues. Off-chain computation, hybrid edge-cloud architectures, and BaaS models should be investigated further.
- **Integration with AI and Federated Learning:** Blockchain integration with AI and federated learning is gaining popularity for privacy-preserving, decentralized model training in pest/disease prediction and crop production forecasting. Future research should look at secure data exchange protocols and model validation frameworks for these hybrid systems.
- **Usability and Adoption:** Despite technology breakthroughs, farmers continue to face significant user experience challenges. Researchers should look into how user-friendly interfaces, education, and policy frameworks might help non-technical stakeholders utilize blockchain technology, particularly in smallholder farming.
- **Environmental Impact and Sustainability:** While blockchain can contribute to PA's environmental goals, it's important to examine its energy footprint. Future research should evaluate the environmental costs and advantages of blockchain deployments in agriculture, as well as investigate strategies to integrate them with green computing projects.
- **Security Beyond Data Integrity:** While most research focuses on data immutability and authentication, other areas such as access control, user identity management, and threat modeling in hostile contexts are underexplored.
- **Cost-benefit analysis:** As farmers and stakeholders consider the adoption of blockchain technology, understanding the economic viability of these systems becomes critical. We plan to extend our previous work [90], by incorporating blockchain technologies into the investment return evaluation framework.

Addressing these directions will bring the scientific community closer to realizing blockchain's full promise for generating sustainable, resilient, and intelligent agricultural systems.

## 12. Conclusions

This literature review systematically examined the integration of blockchain technology in the domain of PA, analyzing 80 peer-reviewed studies published up to August 2024. The analysis reveals a growing body of work exploring blockchain's potential to address core PA challenges, including data integrity, transparency, automation, scalability, and trust.

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By categorizing blockchain-enabled applications according to the agricultural tasks they support—monitoring, control/management, and prediction—we highlighted the diverse use cases of blockchain across different farming contexts. The review also dissected the architectural design of these systems across four layers: physical, edge, service, and application. Notably, Ethereum, Hyperledger, and IOTA emerged as the most commonly used blockchain platforms, often integrated with IoT, AI, and edge computing technologies.

Our findings underscore that while blockchain offers compelling advantages for PA, several technical and practical challenges remain. Issues such as energy efficiency, interoperability, and the complexity of integrating heterogeneous systems limit large-scale adoption. Despite these challenges, the results indicate that blockchain is gradually maturing into a core component of future Agriculture 5.0 infrastructures, enabling more secure, transparent, and autonomous farming ecosystems.

This review not only maps the current landscape but also provides actionable insights into how blockchain contributes to solving domain-specific problems in PA, offering a foundation for both researchers and practitioners seeking to build robust, blockchain-enhanced agricultural systems.

**Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the author(s) used OpenAI's ChatGPT in order to improve the clarity, coherence, and grammar of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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